

PUSH POSSIBLE™

MANHATTAN SCALE™

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

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Contact Address:

Manhattan Associates, Inc.
2300 Windy Ridge Parkway,
Atlanta, Georgia 30339

<http://www.manh.com/>

Region	Telephone	Email
Americas	+1 877.756.7435 (U.S. and Canada only) +1 404.965.4025	callcenter@manh.com
Asia	(00) 800.988.0885 (China only) +86 21 3311 3499	asiacustomersupport@manh.com
Europe, Middle East, and Africa	+44 (0) 1344 318400 (UK) +31 (0)30 214 3400 (NL)	emeacustomersupport@manh.com
Australia	+61 1300787050	aucustomersupport2@manh.com

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SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Revisions

Date	Description	Applies to build
March 29, 2017	Created	14.0.1358.0
November 10, 2017	Added Client secret key and Reply URL	15.0.0168.0
January 01, 2018	Added Register SCALE as Native Application	15.0.0273.0
February 14, 2018	Added Register SCALE as Native Application with AD FS	15.0.0394.0
July 31, 2018	Added URL load balancer	15.0.0394.0
August 09, 2018	Added SCALE application Permission	15.0.0394.0
Nov 6 th 2018	Added Native application scope	16.0.0101.0
May 30 th 2019	Removed native application	16.0.0715.0
Sep 20 th 2019	Modified Set-SingleSignOn PowerShell screenshot	17.0.0065.0
June 13 th 2023	Added config user role	23.0.200.0
June 12 th 2024	Changed Azure AAD to Microsoft Entra ID.	

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Overview

The SCALE metadata web application, RF, web services, and Desktop applications use single sign-on (SSO) to allow users access to SCALE with their standard company credentials. SCALE accomplishes this using the OAuth 2.0 protocol with OpenID Connect. SCALE supports the following SSO services:

- Microsoft Entra ID
- Active Directory Federation Services (AD FS)

This guide is for setting up Single Sign-on with Microsoft Entra ID.

Microsoft Entra ID

Microsoft Entra ID is Microsoft's multi-tenant cloud-based directory and identity management service.

Prerequisites

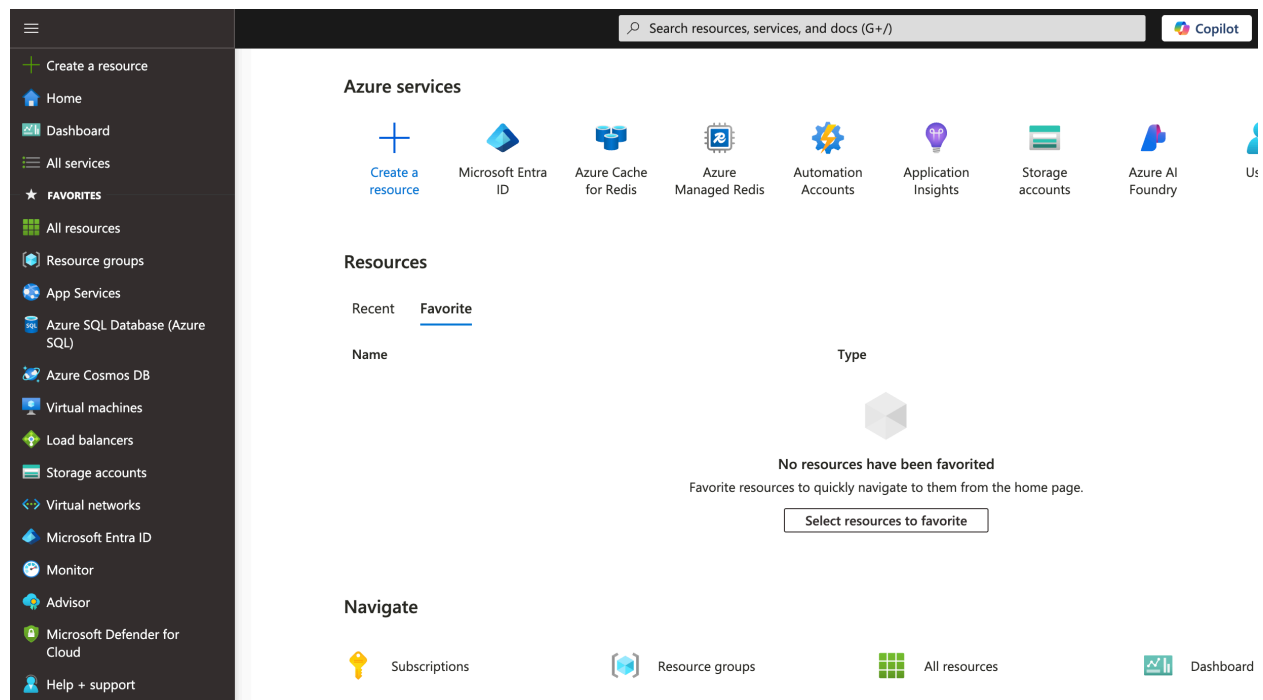
- Configured Microsoft Entra ID.
- Basic understanding of Domain Controller and Microsoft Entra ID.

Register SCALE as Application

For SCALE to work with Microsoft Entra ID, we need to register SCALE as an application in Microsoft Entra ID.

1. Log on to the Microsoft Entra ID portal (<http://aad.portal.azure.com>) directly or log on to the Azure portal (<https://portal.azure.com>) and check if you are on **Microsoft Entra ID** page shown in **Figure 2** if not navigate to **Microsoft Entra ID**.

Figure 1



2. Click **Microsoft Entra ID**.

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Figure 2

Microsoft Azure

Home > **R&D Scale Development | Overview**

Overview | Monitoring | Properties | Recommendations | Setup guides

Search your tenant

Basic information	
Name	R&D Scale Development
Tenant ID	68eedf92-3a9d-4864-9e8a-85edce6c9086
Primary domain	manhdevscale.com
License	Microsoft Entra ID Free

Alerts

- Global Administrators 9
- Migrate to the converged Authentication methods policy

- Click **App registrations** in the menu. Click **New registration** to create a new application.

Figure 3

Home > R&D Scale Development - App registrations

R&D Scale Development - App registrations

Azure Active Directory

Search (Ctrl+/)

+ New registration | Endpoints | Troubleshooting | Got feedback?

Welcome to the new and improved App registrations (now Generally Available). See what's new →

Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)
Still want to use App registrations (Legacy)? [Go back and tell us why](#)

All applications | Owned applications

Start typing a name or Application ID to filter these results

DISPLAY NAME	APPLICATION
	This account i

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4. On the **Register an application** screen, enter a name for the application in the **Name** field. Select supported account types as **“Account in the organization directory only”**.

Figure 4

Home > R&D Scale Development > App registrations

Register an application

*** Name**
The user-facing display name for this application (this can be changed later).

Manhattan SCALE 2019 ✓

Supported account types
Choose the account types that can use this application or access this API

Single tenant only - R&D Scale Development Help me choose

Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Public client/native (mobile ... e.g. myapp://auth ✓

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

By proceeding, you agree to the [Microsoft Platform Policies](#)

Register

5. Enter the application URL (i.e. *https://<scale server full machine name> OR <DNS of load Balancer pointing to SCALE>/scale*) in the **Redirect-URI** field and select **Web**. Click **Register**.

Figure 5

Home > R&D Scale Development > App registrations

Register an application

*** Name**
The user-facing display name for this application (this can be changed later).

Manhattan SCALE 2019 ✓

Supported account types
Choose the account types that can use this application or access this API

Single tenant only - R&D Scale Development Help me choose

Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Web https://test.sampledomain.com/scale ✓

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

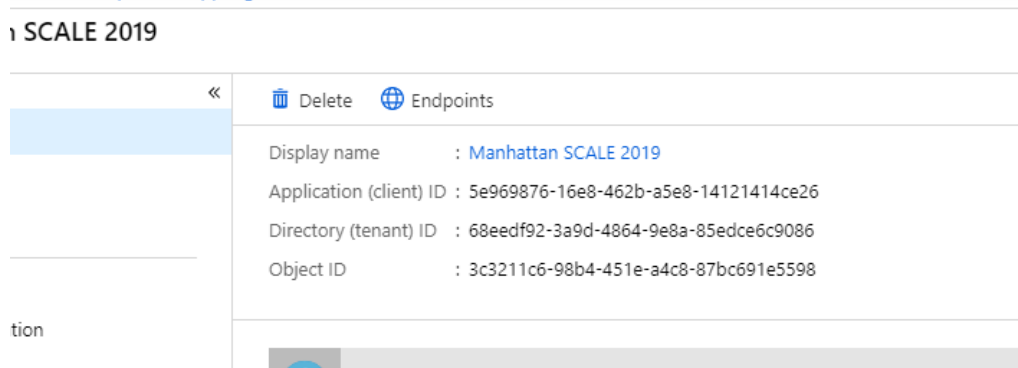
By proceeding, you agree to the [Microsoft Platform Policies](#)

Register

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

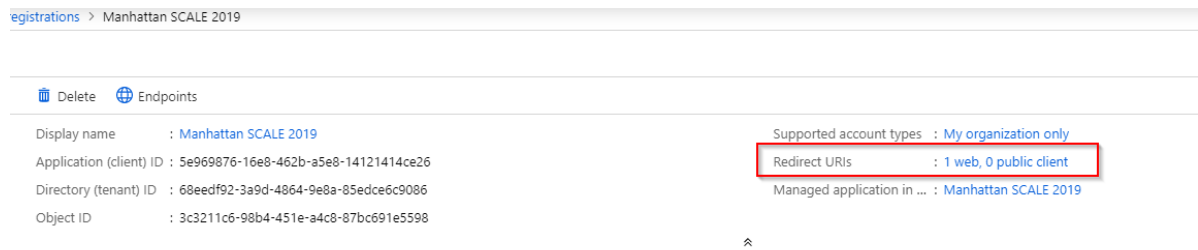
- Copy the **Application ID** and **Directory ID** for the application after it has been created. Later you will configure the SCALE application to use this.

Figure 6



- Click **Redirect URIs**.

Figure 7



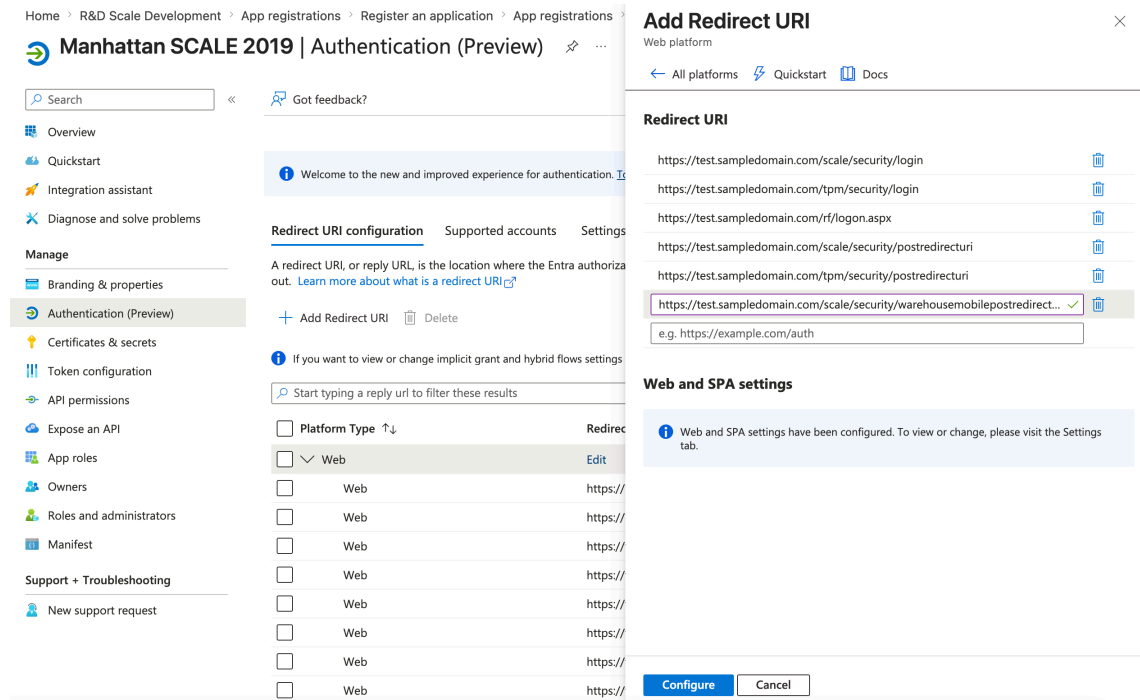
- This step is only for customers without any SCALE environments configured.** Click **Add Redirect URI** and then configure the 6 **Web** reply URLs. Click **Configure**.

Note: hostname is fully qualified domain name pointing to SCALE server/s. To access SCALE endpoints (RF, SCALE, Web Service, TPM etc.) use this hostname. In below Example H-dalal-16.ilsnetdev.pri is hostname pointing to SCALE server. To access SCALE/RF/TPM/Web Services use URL <https://<hostname>/Scale/> or <https://<hostname>/rf/logon.aspx> or <https://<hostname>/TPM>. Configure the same hostname as URL load balancer in SCALE toolbox as discussed [below](#).

- <https://<hostname>/scale/security/login>
- <https://<hostname>/tpm/security/login>
- <https://<hostname>/rf/logon.aspx>
- <https://<hostname>/scale/security/postredirecturi>
- <https://<hostname>/tpm/security/postredirecturi>
- <https://<hostname>/scale/security/warehousemobilepostredirecturi/>

Figure 8

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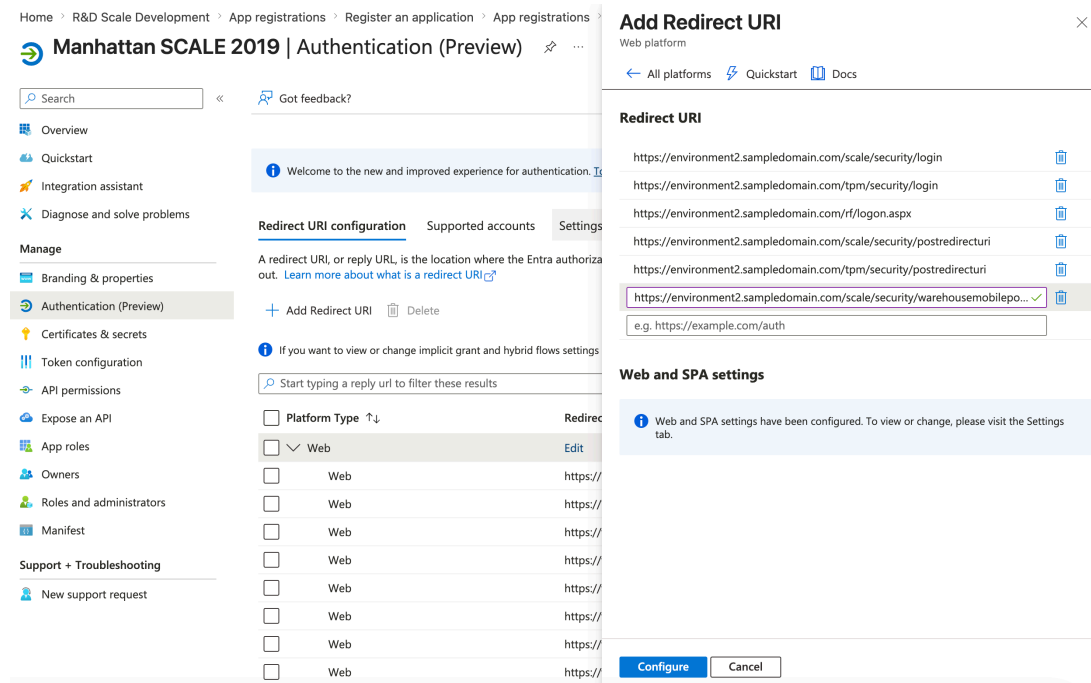
9. **This step is only for customers with 1 or more SCALE environments configured.** Click **Add Redirect URI** and configure 6 **Web** reply URLs per environment. Click **Configure**.

Note: hostname is the fully qualified domain name pointing to SCALE server/s. Subdomain of the hostname should be the name of the SCALE environment. To access SCALE endpoints (RF, SCALE, Web Service, TPM etc.) use this hostname. In the below Example environment1.ilnetdev.pri is the hostname pointing to the SCALE server with environment1 configured as the SCALE environment. To access SCALE/RF/ TPM/Web Services use URL `https://<hostname>/Scale` or `https://<hostname>/rf/logon.aspx` or `https://<hostname>/TPM`. Configure the same hostname as the URL load balancer in the SCALE toolbox as discussed below.

- `https://< hostname >/scale/security/login`
- `https://< hostname >/rf/logon.aspx`
- `https://< hostname >/scale/security/postredirecturi`
- `https://< hostname >/tpm/security/postredirecturi`
- `https://< hostname >/tpm/security/login`
- `https://<hostname>/scale/security/warehousemobilepostredirecturi/`

Figure 9

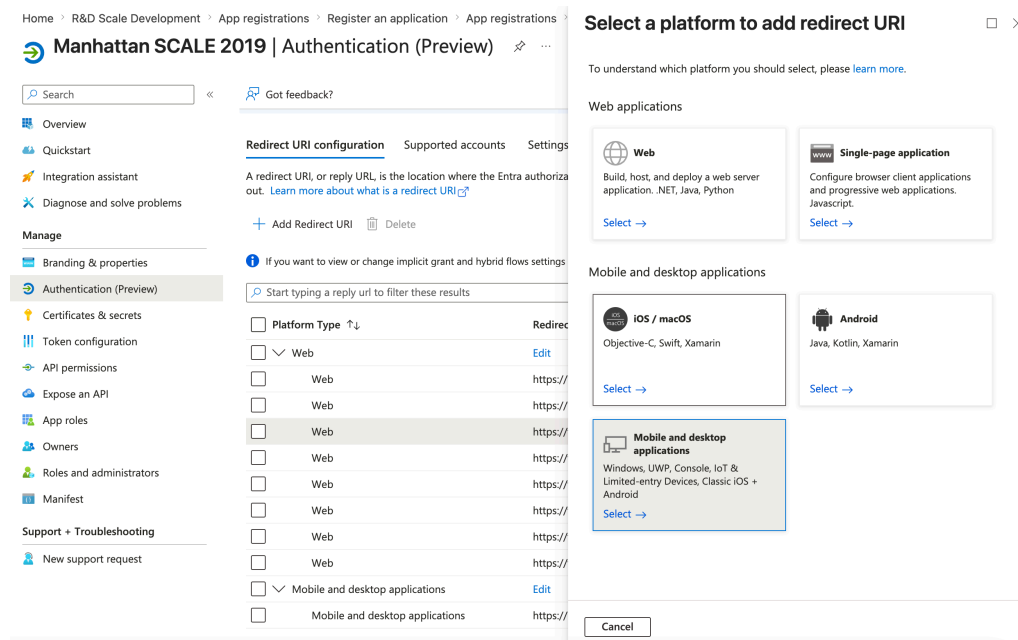
SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



10. Check below **suggested Redirect URIs for public clients** on the same screen (mobile, desktop).
Follow the steps below

Click **+ Add Redirect URI** button
select **Mobile and desktop applications**

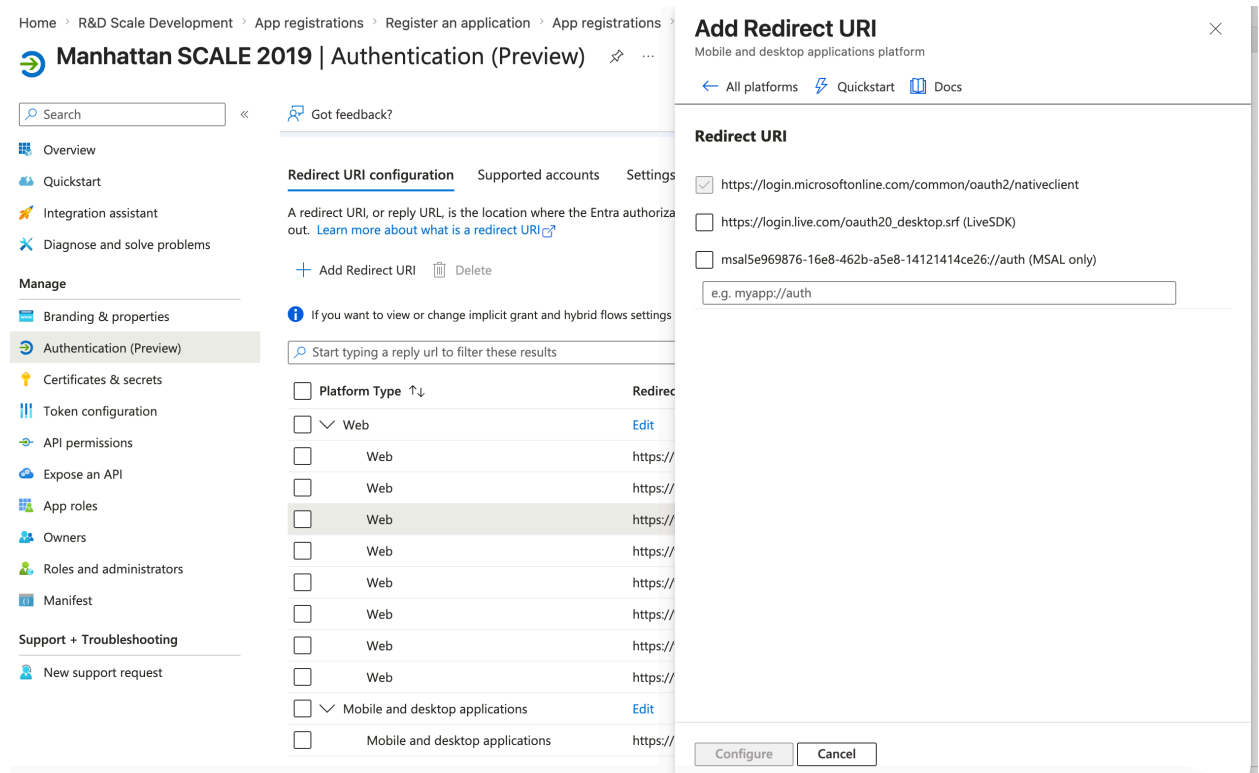
Figure 10.1



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Select the check the box <https://login.microsoftonline.com/common/oauth2/nativeclient> as shown below.

Figure 10.2



11. Navigate to **Settings** tab on the same screen. Click **Save**

Figure 11

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Home > R&D Scale Development > App registrations > Register an application > App registrations > Manhattan SCALE 2019

Manhattan SCALE 2019 | Authentication (Preview)

Search Got feedback?

- Overview
- Quickstart
- Integration assistant
- Diagnose and solve problems

Manage

- Branding & properties
- Authentication (Preview)**
 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
 - App roles
 - Owners
 - Roles and administrators
 - Manifest
- Support + Troubleshooting
 - New support request

Welcome to the new and improved experience for authentication. [To switch to the old experience, please click here.](#)

Redirect URI configuration Supported accounts **Settings**

Web and SPA settings

The following settings are only applicable when building a web or SPA application.

Implicit grant and hybrid flows

The implicit grant allows an application to request a token directly from the authorization endpoint. It is only recommended for browser-based single-page applications (SPAs).

☐ Access tokens (used for implicit flows)

☒ ID tokens (used for implicit and hybrid flows)

Front-channel logout URL

e.g. <https://example.com/logout>

Allow public client flows

[Learn more](#)

☐ Disabled

App instance property lock

[Learn more](#)

Configure the application instance modification lock. [Configure](#)

[Save](#) [Discard](#)

12. Click **App roles** on the left. Create the following app roles.

Figure 12

Display name	Description	Allowed member types	Value
CONFIGUSER	CONFIGUSER	Users/Groups	CONFIGUSER
Confidential	Confidential	Applications	Confidential
TPMUSER	TPMUSER	Users/Groups	TPMUSER
Administrators	Administrators	Users/Groups	Administrators
SCALEUSER	SCALEUSER	Users/Groups,Applications	SCALEUSER

13. Navigate back to the **Microsoft Entra ID**. Click **Enterprise applications** and select the application created above.

Figure 13

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

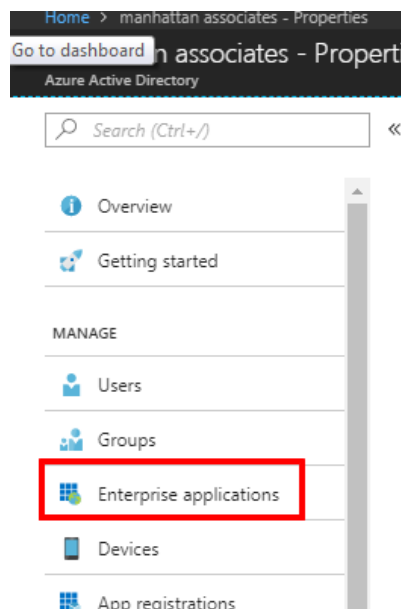
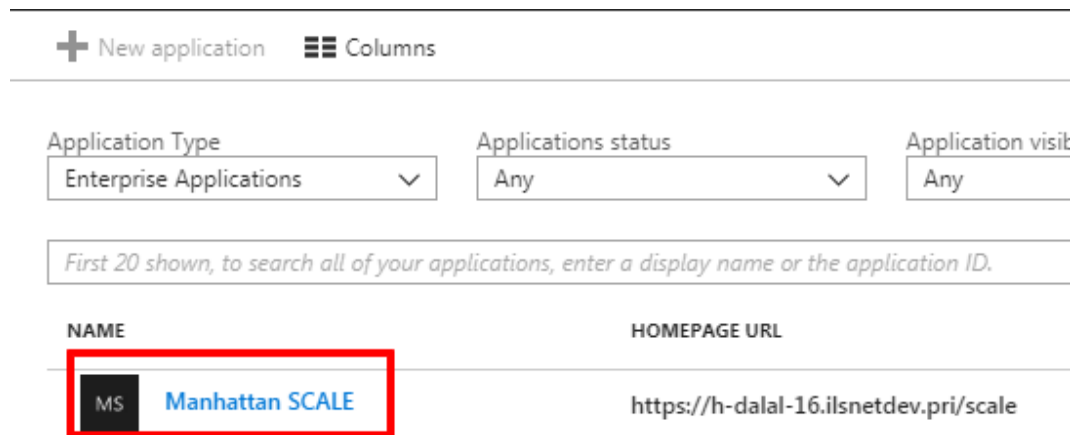


Figure 17



14. Click **Users and groups** in the menu and then click **Add user**.

Figure 18

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

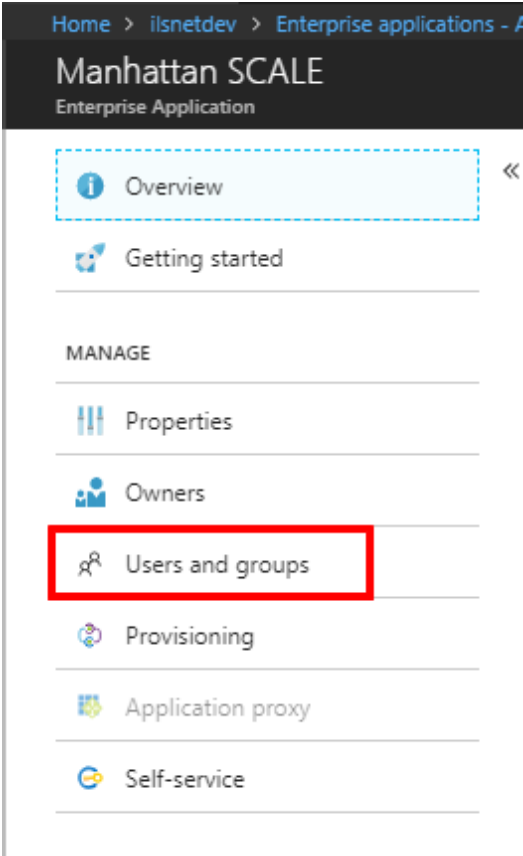
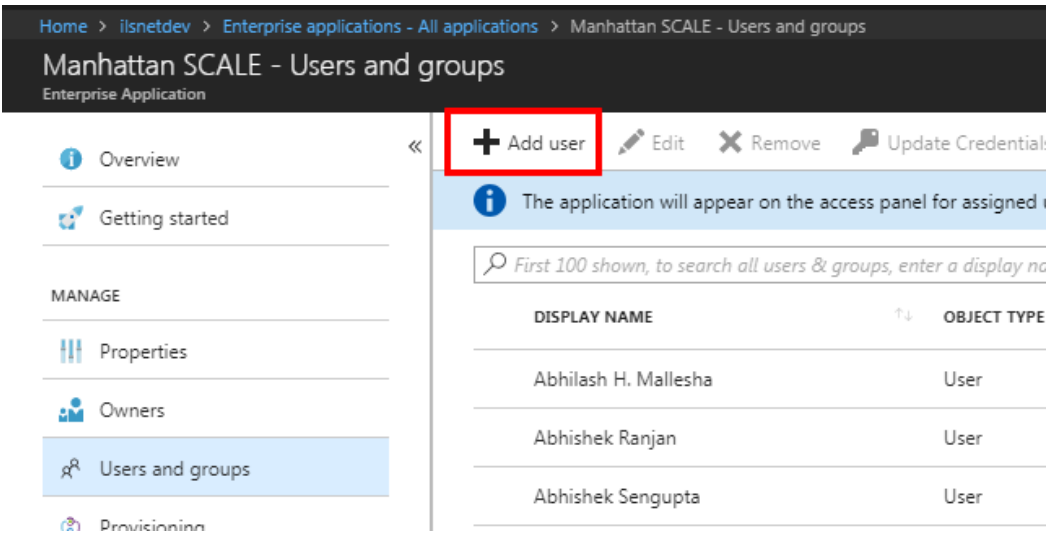


Figure 19

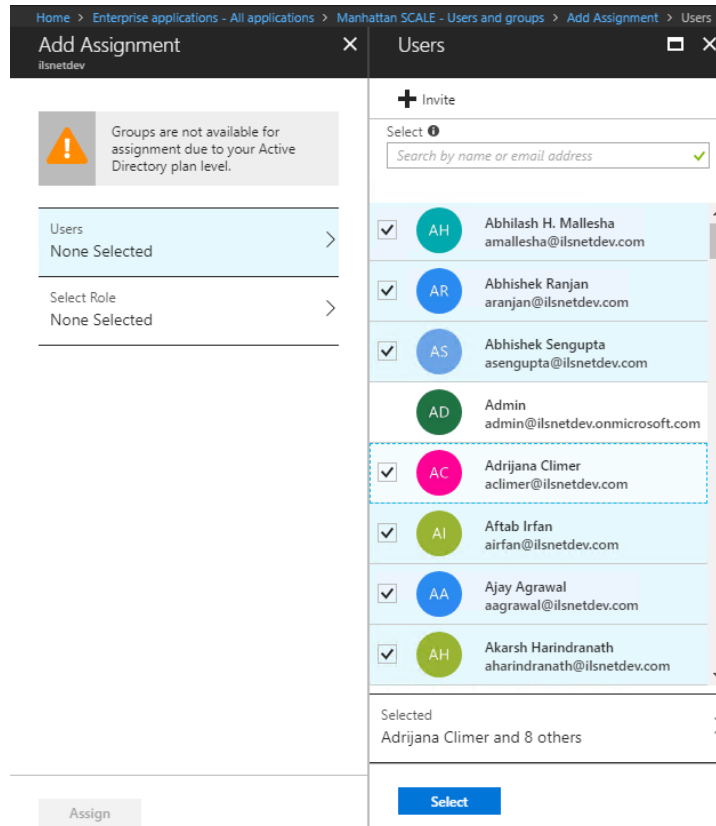


15. Click **Users** and select the users that need access to the SCALE application.

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Note: If you have Basic plan or higher of Microsoft Entra ID you can add a group and assign roles to it.

Figure 20

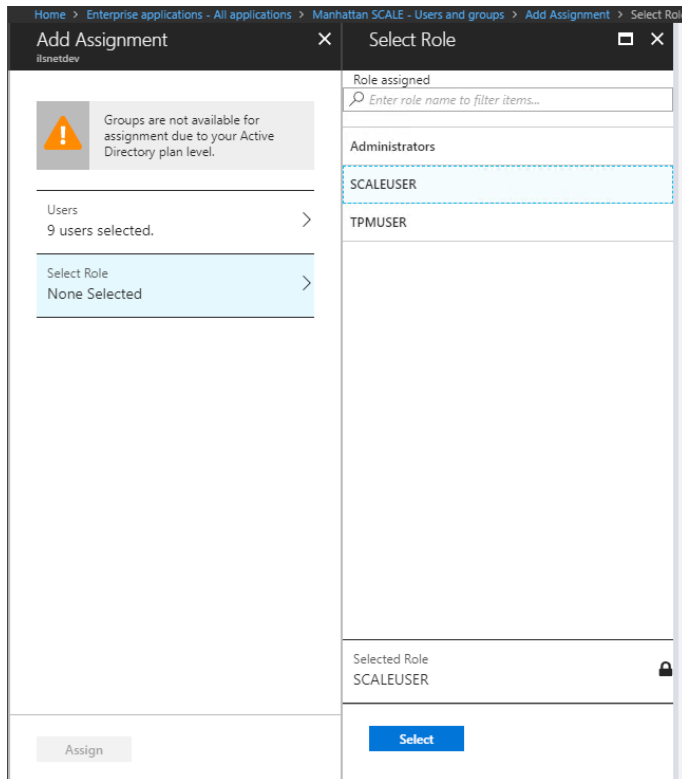


16. Click **Select Role** and choose the role for the users.

- SCALEUSER – This role gives access to SCALE web app, RF, Configuration API's and TPM.
- CONFIGUSER – This role gives access to Config API used in the Configuration web app.
- TPMUSER - This role gives access to TPM only.
- Administrators - This role is for future use.

Figure 21

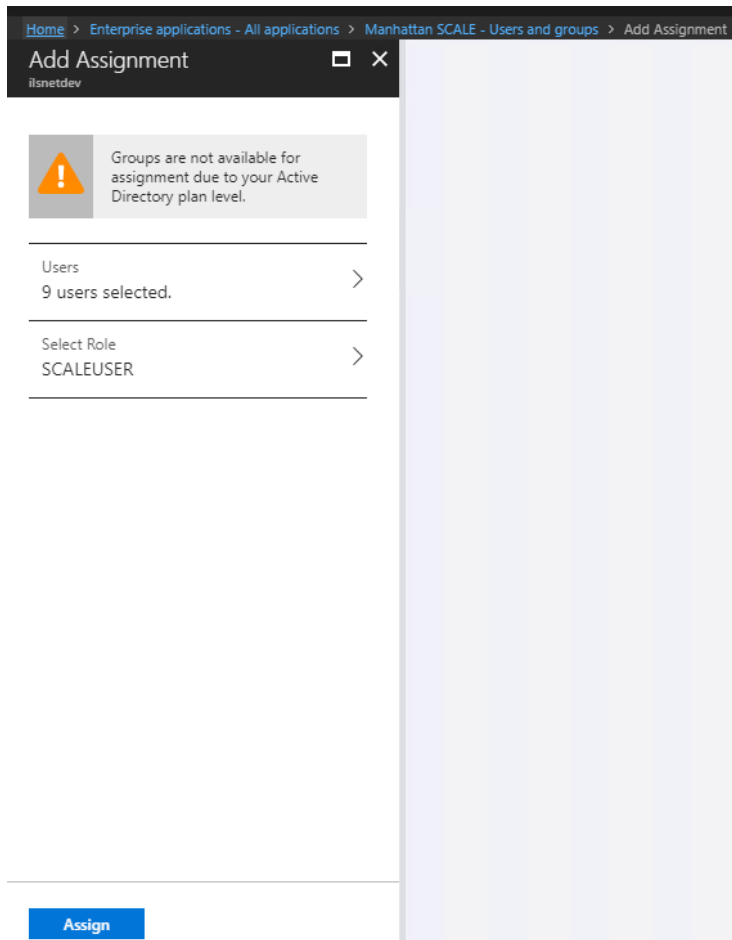
SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



17. Click **Assign**.

Figure 22

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



18. The selected users will be displayed with their assigned roles on the next screen.

Figure 23

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

+ Add user **Edit** **Remove** **Update Credentials**

i The application will appear on the access panel for assigned users. Set 'visible to users?' to no in properties to prevent

First 100 shown, to search all users & groups, enter a display name.

DISPLAY NAME	OBJECT TYPE	ROLE ASSIGNED
Yan Fu	User	SCALEUSER
Abhishek Sengupta	User	SCALEUSER
Aftab Irfan	User	SCALEUSER
Abhishek Ranjan	User	SCALEUSER
Abhilash H. Mallesha	User	SCALEUSER
Swale	User	SCALEUSER
Adrijana Climer	User	SCALEUSER
Ajay Agrawal	User	SCALEUSER
Akarsh Harindranath	User	SCALEUSER

19. Click **Properties** in the menu.

Figure 24

Manhattan SCALE - Properties
Enterprise Application

Overview **Getting started**

MANAGE

- Properties**
- Owners
- Users and groups
- Provisioning

Save **Discard**

Enabled for users to sign-in? **Yes** **No**

Name **i** Manhattan SCALE

Publisher **i** ilsnetdev

Homepage URL **i** https://h-dalal-16.ilsnetdev.pri/scale

20. Switch the **User assignment required?** option to Yes.

Figure 25

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Save Discard

Enabled for users to sign-in? ☒ Yes ☐ No

Name ⓘ Manhattan SCALE

Publisher ⓘ ilsnetdev

Homepage URL ⓘ https://h-dalal-16.ilsnetdev.pri/scale

Logo ⓘ  Select a file

Application ID ⓘ 

Object ID ⓘ 

User assignment required? ⓘ ☒ Yes ☐ No

Visible to users? ⓘ ☒ Yes ☐ No

21. Click **Save**.

Figure 26

Save Discard Delete

Enabled for users to sign-in? ⓘ ☒ Yes ☐ No

Name ⓘ Manhattan SCALE 2019 ✓

Homepage URL ⓘ

Logo ⓘ  Select a file 

Application ID ⓘ 5e969876-16e8-462b-a5e8-14121414ce26 

Object ID ⓘ 39308ef5-0f2b-4871-904f-bc5889821da8 

User assignment required? ⓘ ☒ Yes ☐ No

Visible to users? ⓘ ☒ Yes ☐ No

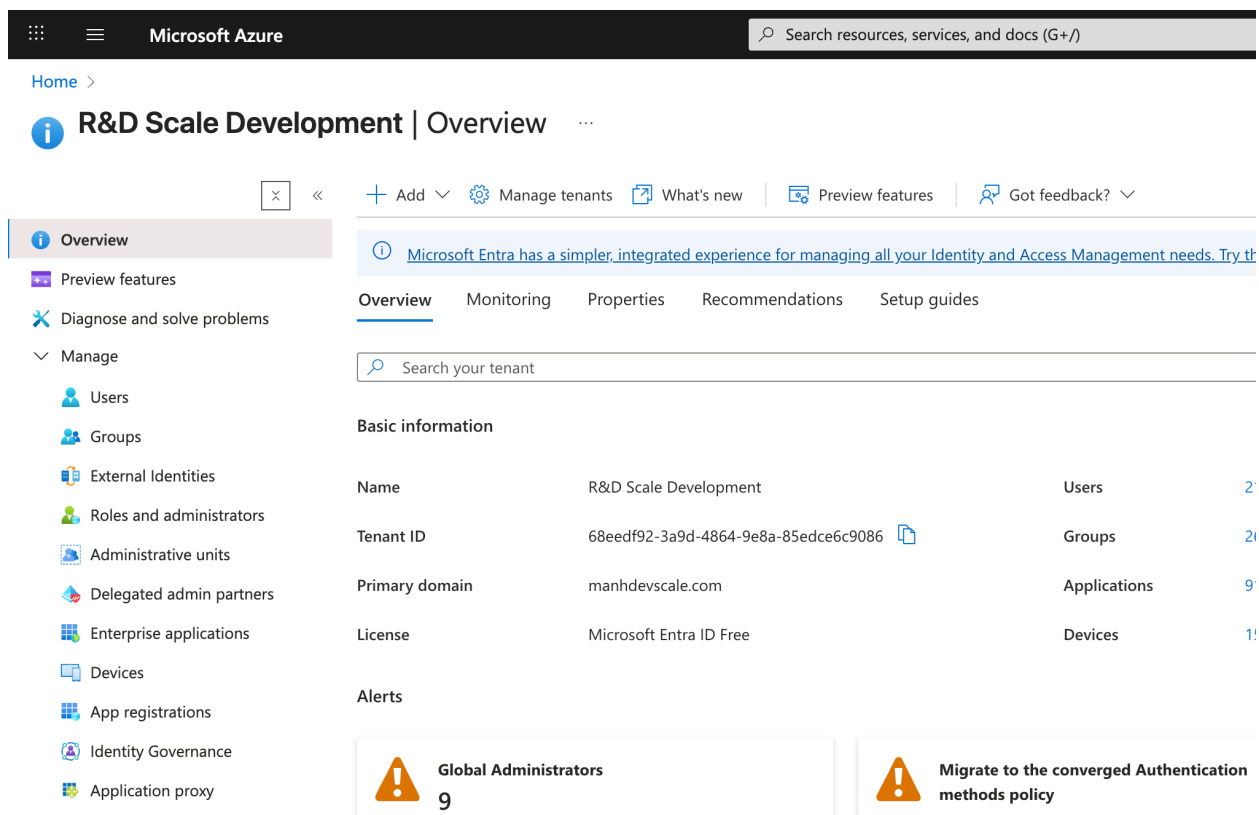
Confidential Application

This is OAuth 2.0 client credentials grant flow, to access web-hosted resources by using the identity of an application. This type of grant commonly is used for server-to-server interactions. These types of applications often are referred to as daemons. Service accounts require a client secret key to be supplied.

Below are the steps to create a confidential application.

1. Log on to the Azure AAD portal (<http://aad.portal.azure.com>) directly or log on to the Azure portal (<https://portal.azure.com>) and navigate to **Microsoft Entra ID**.

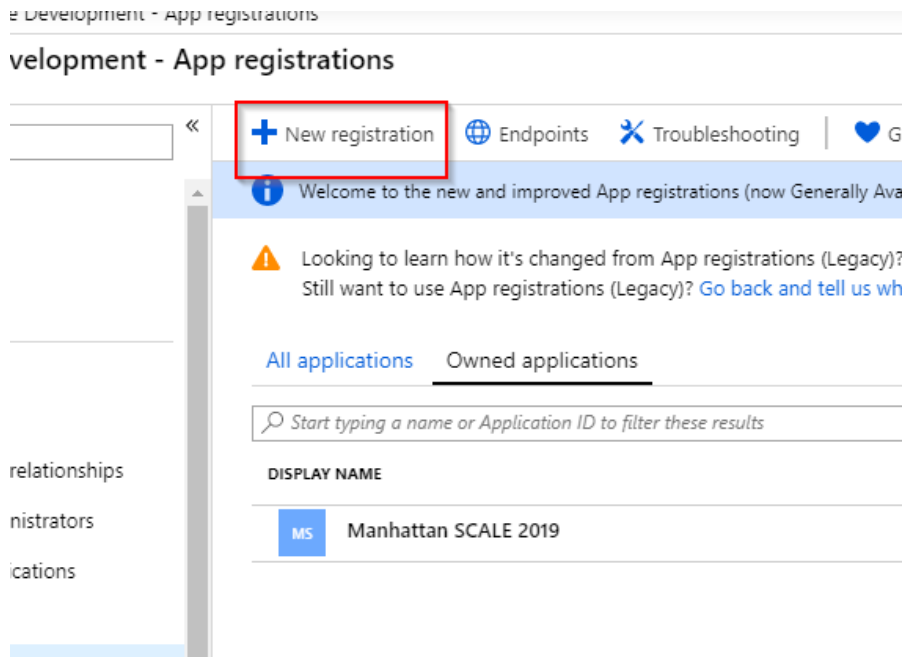
Figure 27



2. Select your directory, and then click **App registrations** in the menu. Click **New registration** to create a new application.

Figure 28

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



3. On the **Register an application** screen, enter a name for the application in the **Name** field. Click **Register**

Figure 29

Home > Enterprise applications | All applications > Manhattan SCALE 2019 | Overview > App registrations

Register an application

*** Name**
The user-facing display name for this application (this can be changed later).

Supported account types
Choose the account types that can use this application or access this API
 [Help me choose](#)

Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

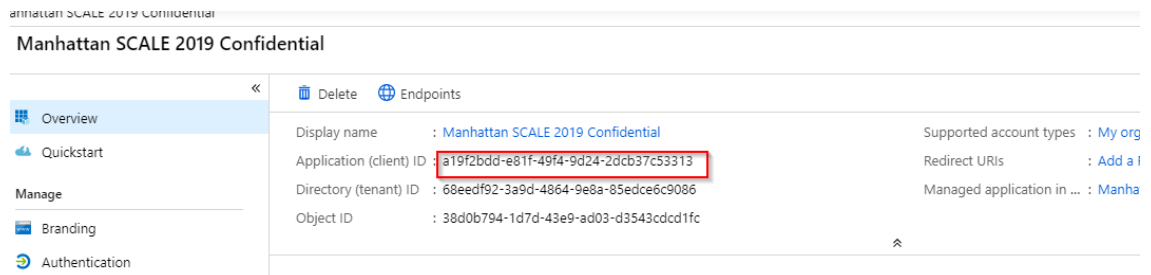
Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

By proceeding, you agree to the [Microsoft Platform Policies](#)

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

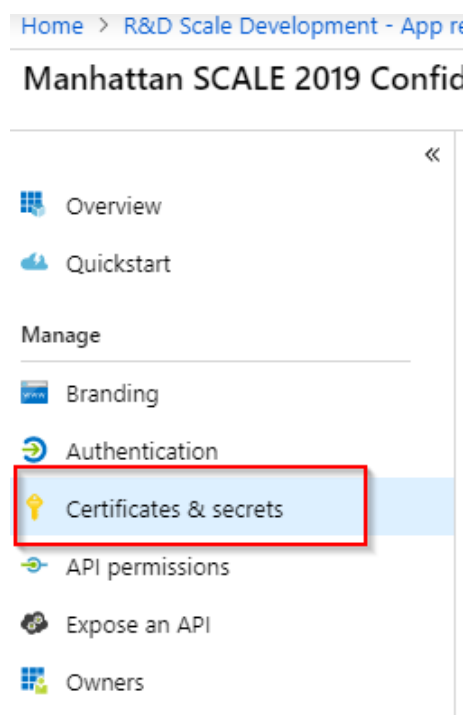
4. Note the **Application ID** for the application after it has been created. You will configure the SCALE application to use this later.

Figure 31



5. Click **Certificates & secrets**.

Figure 32



6. Click **New client secret**.

Figure 33

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Client secrets

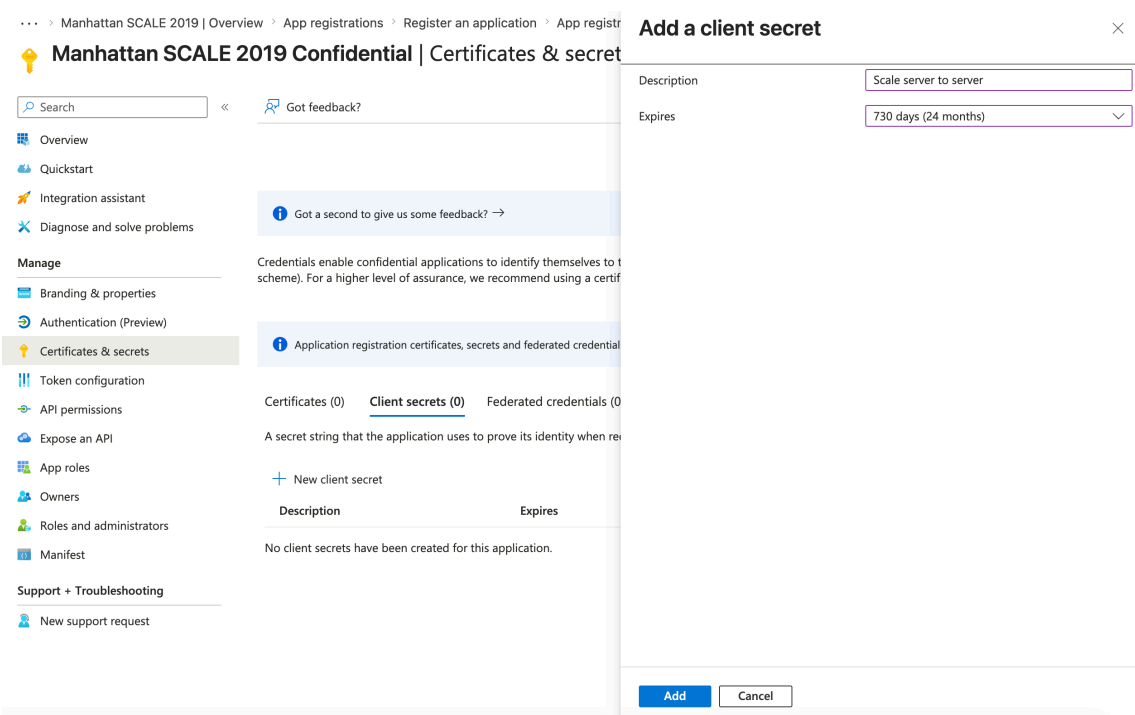
A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

[+ New client secret](#)

DESCRIPTION	EXPIRES	VALUE
No client secrets have been created for this application.		

7. Enter a **Key Description** and select a **Duration**. Click **Add**.

Figure 34



8. The **secret** will be displayed when you click Add. Make a note of this key as it will be required for further steps. Once you exit the screen, the key will not be visible.

Figure 35

Certificates (0)

Client secrets (1)

Federated credentials (0)

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

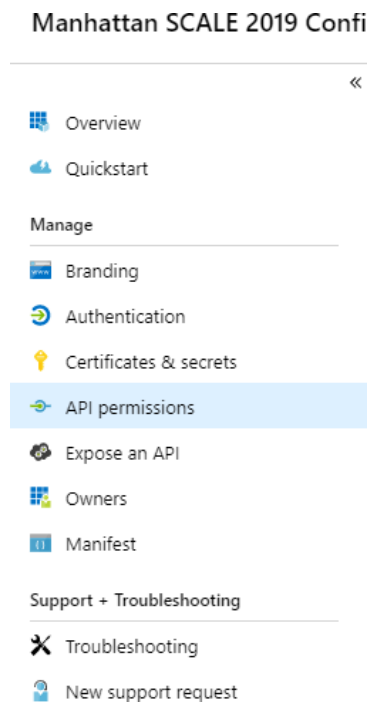
+ New client secret

Description	Expires	Value ⓘ	Secret ID
Scale server to server	6/18/2028	2Oa8Q~hgJebkep~kxN2.rolheZgBO~kk... ⓘ	e5513437-3ad9-42f7-85c9-c41110594bd6 ⓘ ⓘ

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

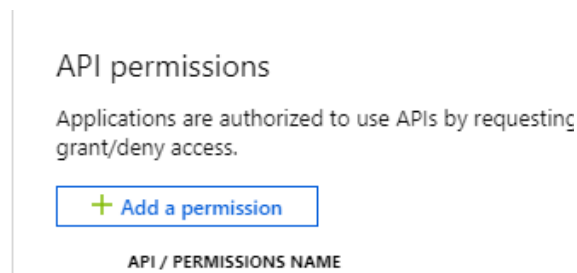
9. Click **API permissions** in the Settings menu.

Figure 36



10. Click **Add a permission**.

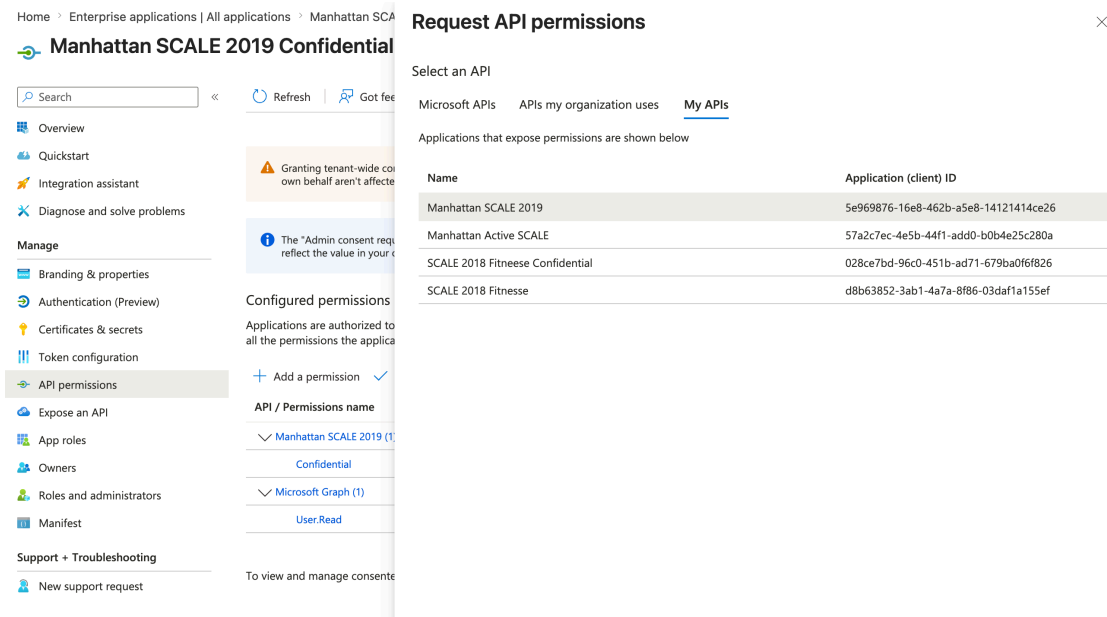
Figure 37



11. In the Select an API access dialog, click **Select My APIs**.

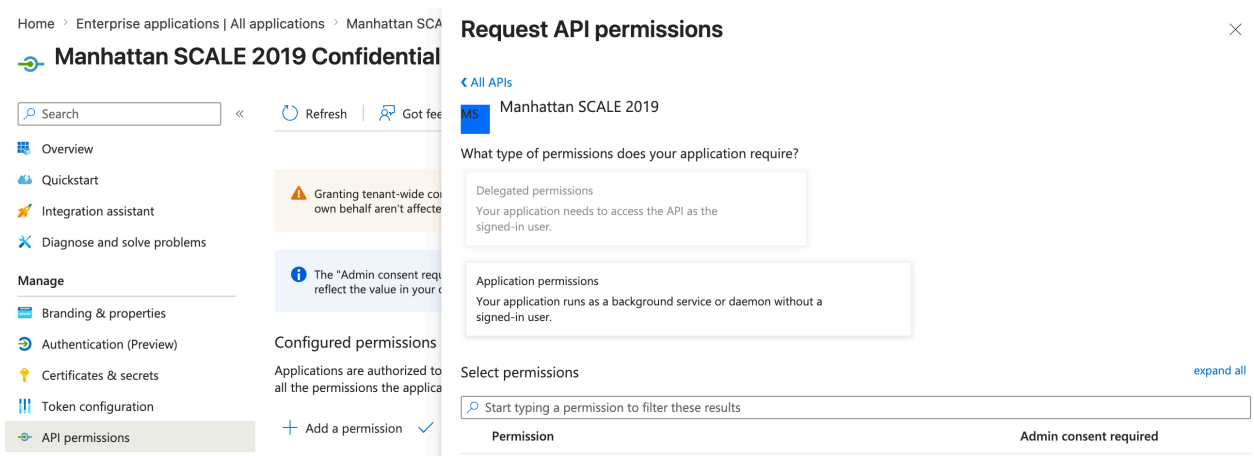
Figure 38

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12. Select the SCALE application created in the [Register SCALE as Application](#) section and click the application.

Figure 39



13. Select the **Confidential** checkbox and click **Add permissions**.

Figure 40

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Request API permissions

< All APIs

Manhattan SCALE 2019

What type of permissions does your application require?

Delegated permissions
Your application needs to access the API as the signed-in user.

Application permissions
Your application runs as a background service or daemon without a signed-in user.

Select permissions [expand all](#)

Start typing a permission to filter these results

Permission	Admin consent required
Permissions (1)	
☒ Confidential Confidential	Yes
☐ SCALEUSER SCALEUSER	Yes

Add permissions Discard

14. Click **Grant admin consent for directory name** and click **Yes**.

Figure 42

Permissions have changed. Users and/or admins will have to consent even if they have already done so previously.

API permissions

Applications are authorized to use APIs by requesting permissions. These permissions show up during the consent process where users are given the opportunity to grant/deny access.

[+ Add a permission](#)

API / PERMISSIONS NAME	TYPE	DESCRIPTION	ADMIN CONSENT REQUIRED
Manhattan SCALE 2019 (1)			
Confidential	Application	Confidential	Yes ⚠ Not granted for R&D ...
Microsoft Graph (1)			
User.Read	Delegated	Sign in and read user profile	-

These are the permissions that this application requests statically. You may also request user consent-able permissions dynamically through code. [See best practices for requesting permissions](#)

Grant consent

As an administrator, you can grant consent on behalf of all users in this directory. Granting admin consent for all users means that end users will not be shown a consent screen when using the application.

[Grant admin consent for R&D Scale Development](#)

Do you want to grant consent for the requested permissions for all accounts in R&D Scale Development? This will update any existing admin consent records this application already has to match what is listed below.

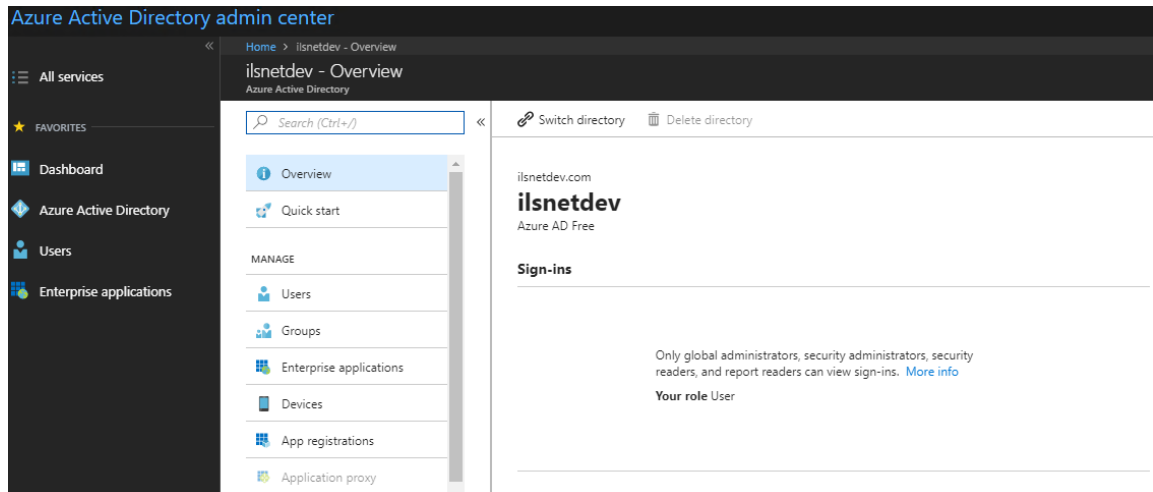
[Yes](#) [No](#)

Note: Only a global administrator can grant this permission.

15. Go to the home page by changing the browser URL to <http://aad.portal.azure.com> and clicking **Enterprise applications**.

Figure 43

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



16. Select the application created above.

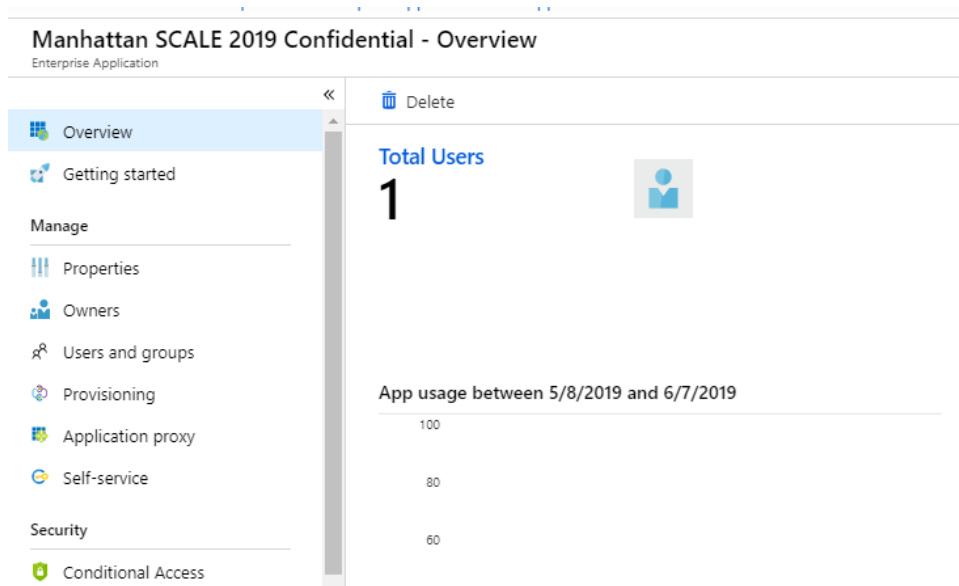
Figure 44

pplications



17. Click **Properties**.

Figure 45



18. Switch the **User assignment required?** option to Yes and click **Save**.

Figure 46

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

The screenshot shows the 'SCALEConfidentialAPP - Properties' page in the Microsoft Entra ID portal. The left sidebar contains navigation links for Overview, Getting started, MANAGE (Properties, Owners, Users and groups, Provisioning, Application proxy, Self-service), SECURITY (Conditional access, Permissions), ACTIVITY (Sign-ins, Audit logs), and TROUBLESHOOTING + SUPPORT. The main content area has 'Save' and 'Discard' buttons at the top. Below them are several settings: 'Enabled for users to sign-in?' (Yes/No), 'Name' (SCALEConfidentialAPP), 'Publisher' (ilsnetdev), 'Homepage URL' (https://h-dalal-16.ilsnetdev.pri/scale), 'Logo' (a black square with 'SC' and a 'Select a file' button), 'Application ID' (22f968e4-7d20-4d55-a1d6-7b39226d549f), 'Object ID' (1b996e91-b0c6-423e-a4e3-bcf0b0c7052d), 'User assignment required?' (highlighted with a red box, Yes/No), and 'Visible to users?' (Yes/No).

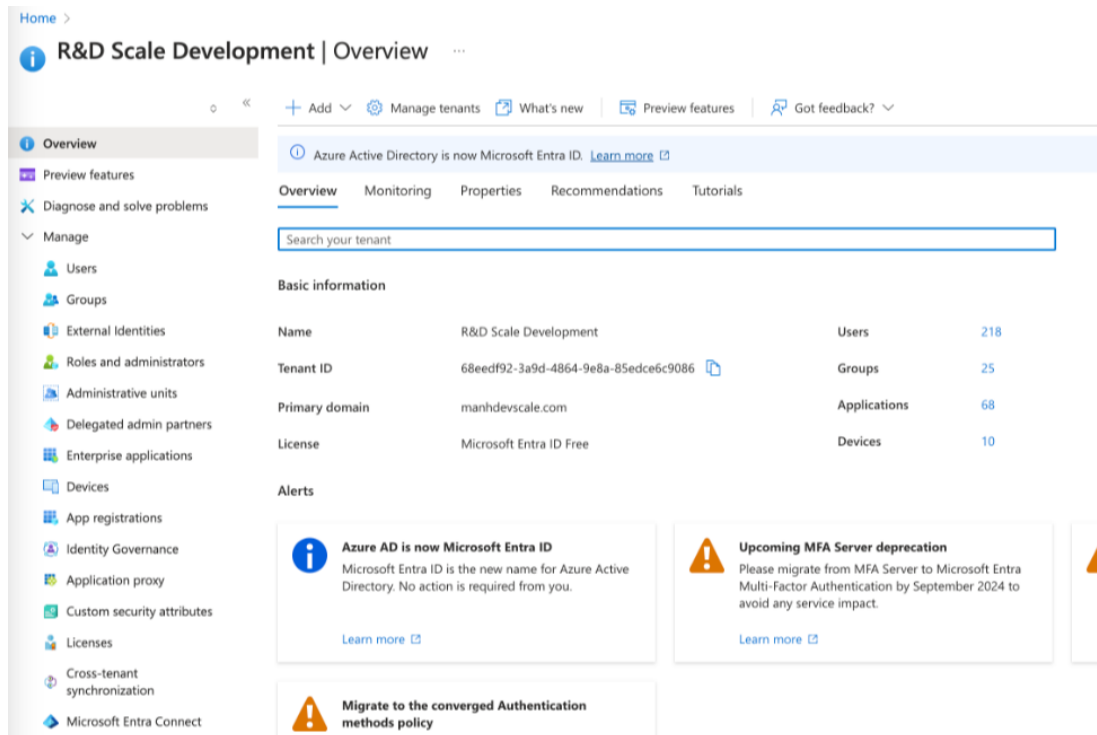
API Confidential Application [Optional]

This is an optional application only required if custom code like MHE, exit points, Azure functions, host system, etc. needs to call SCALE API marked as SCALEUser role. This is OAuth 2.0 client credentials grant flow. The SCALEAPI user needs to be manually created in the SCALE User Profile configurations to make use of this flow. Below are the steps to create this application.

1. Log on to the Azure AAD portal (<http://aad.portal.azure.com>) directly or log on to the Azure portal (<https://portal.azure.com>) and navigate to **Active Directory**.

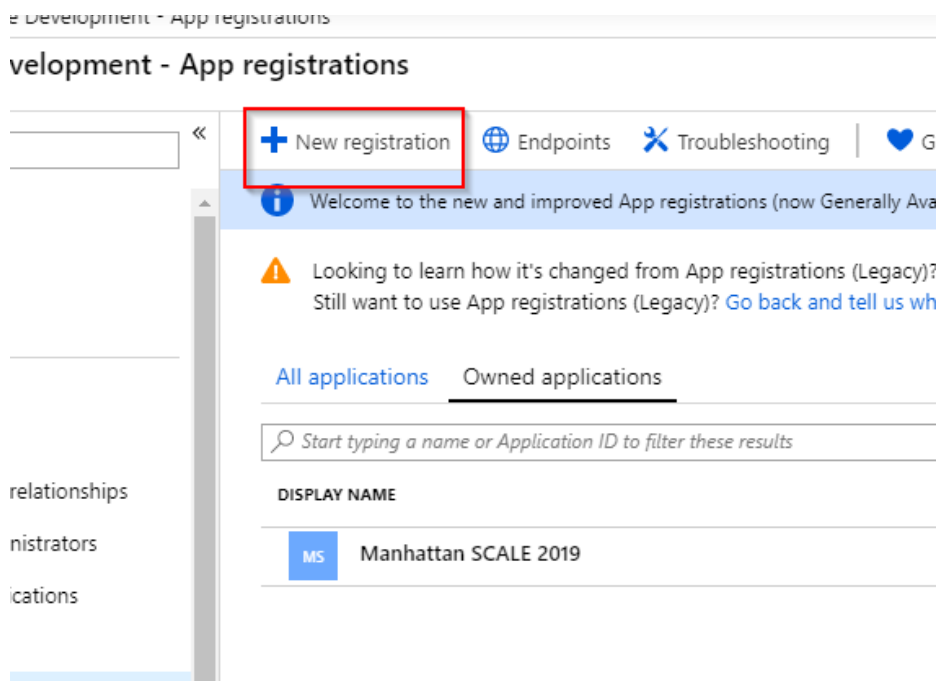
Figure 27

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



19. Select your directory, and then click **App registrations** in the menu. Click **New registration** to create a new application.

Figure 28



20. On the **Register an application** screen, enter a name for the application in the **Name** field. Click **Register**

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Figure 29

Home > Enterprise applications | All applications > Manhattan SCALE 2019 | Overview > App registrations > Register an application > App registrations

Register an application

*** Name**
The user-facing display name for this application (this can be changed later).

SCALE API Confidential ✓

Supported account types
Choose the account types that can use this application or access this API

Single tenant only - R&D Scale Development ▾ [Help me choose](#)

Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform ▾ e.g. https://example.com/auth

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

By proceeding, you agree to the [Microsoft Platform Policies](#)

Register

21. Note the **Application ID** for the application after it has been created. You will configure the SCALE application to use this later.

Figure 31

Manhattan SCALE 2019 Confidential

Manhattan SCALE 2019 Confidential

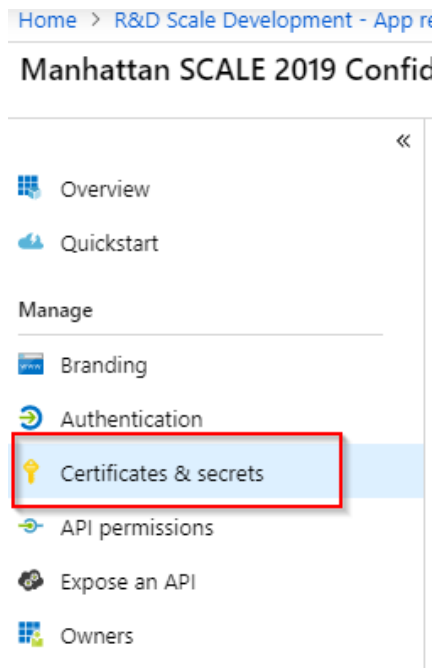
« Delete Endpoints

Overview	Display name : Manhattan SCALE 2019 Confidential	Supported account types : My org
Quickstart	Application (client) ID : a19f2bdd-e81f-49f4-9d24-2dcb37c53313	Redirect URIs : Add a l
Manage	Directory (tenant) ID : 68eedf92-3a9d-4864-9e8a-85edce6c9086	Managed application in ... : Manha
Branding	Object ID : 38d0b794-1d7d-43e9-ad03-d3543cdcd1fc	
Authentication		

22. Click **Certificates & secrets**.

Figure 32

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



23. Click **New client secret**.

Figure 33

Client secrets

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

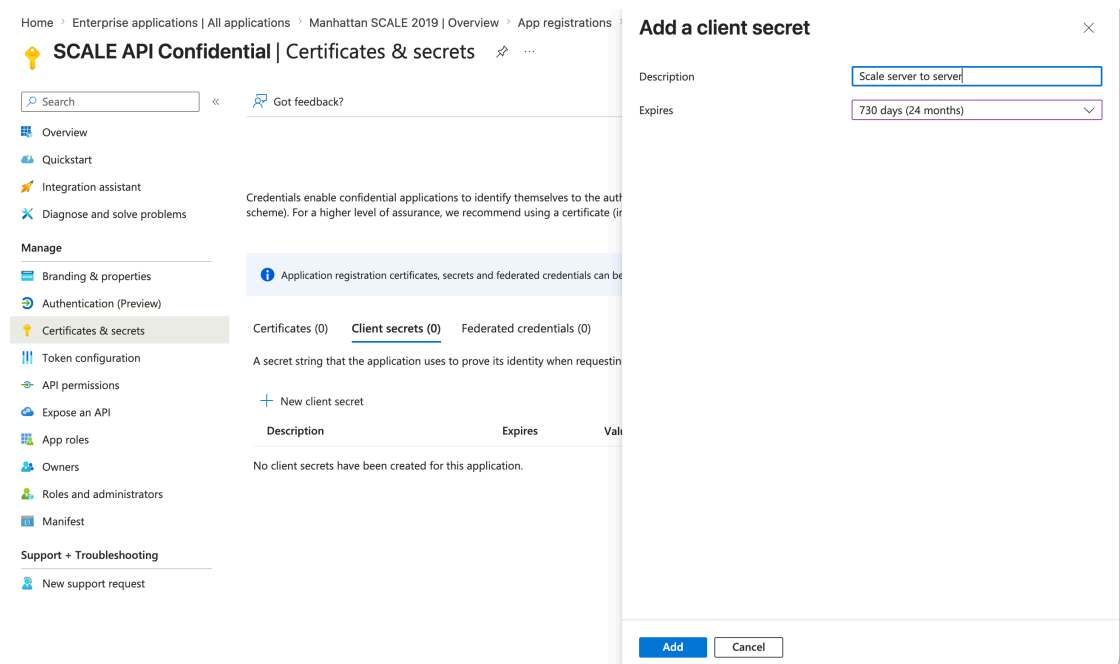
[+ New client secret](#)

DESCRIPTION	EXPIRES	VALUE
No client secrets have been created for this application.		

24. Enter a **Key Description** and select a **Duration**. Click **Add**.

Figure 34

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



25. The secret will be displayed when you click Add. Make a note of this key as it will be required for further steps. Once you exit the screen, the key will not be visible.

Figure 35

Client secrets

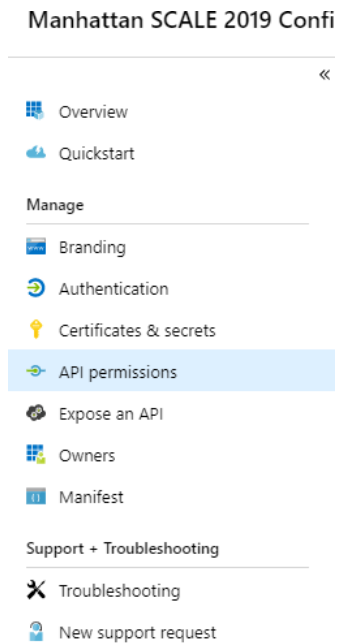
A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

+ New client secret		
DESCRIPTION	EXPIRES	VALUE
SCALE server to server	6/7/2020	tRF4xHIY6@u8Vma1g?-BXk--HPXE+0ns 

26. Click **API permissions** in the Settings menu.

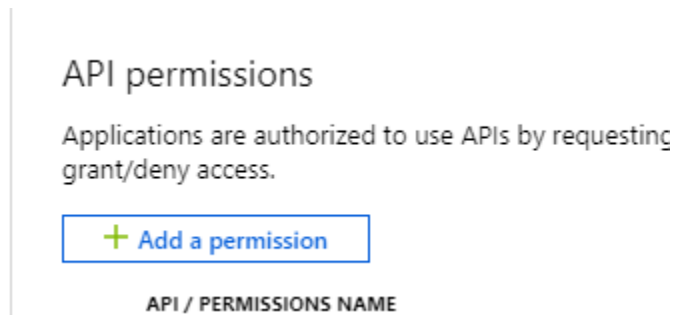
Figure 36

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



27. Click **Add a permission**.

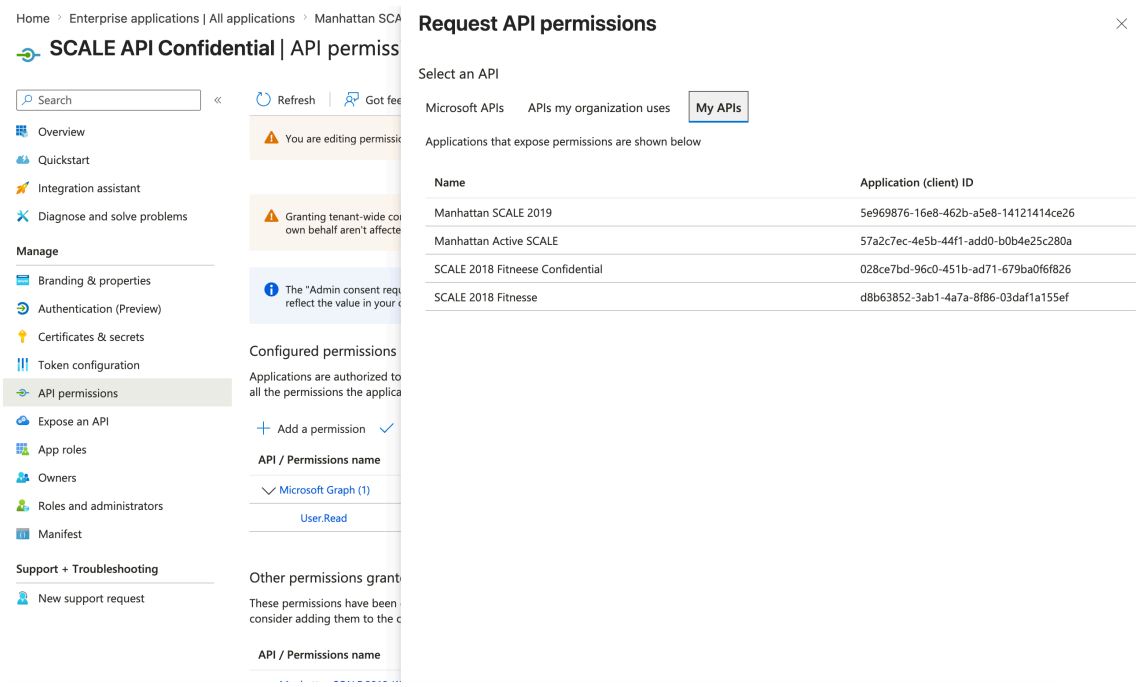
Figure 37



28. In the Select an API access dialog, click **Select My APIs**.

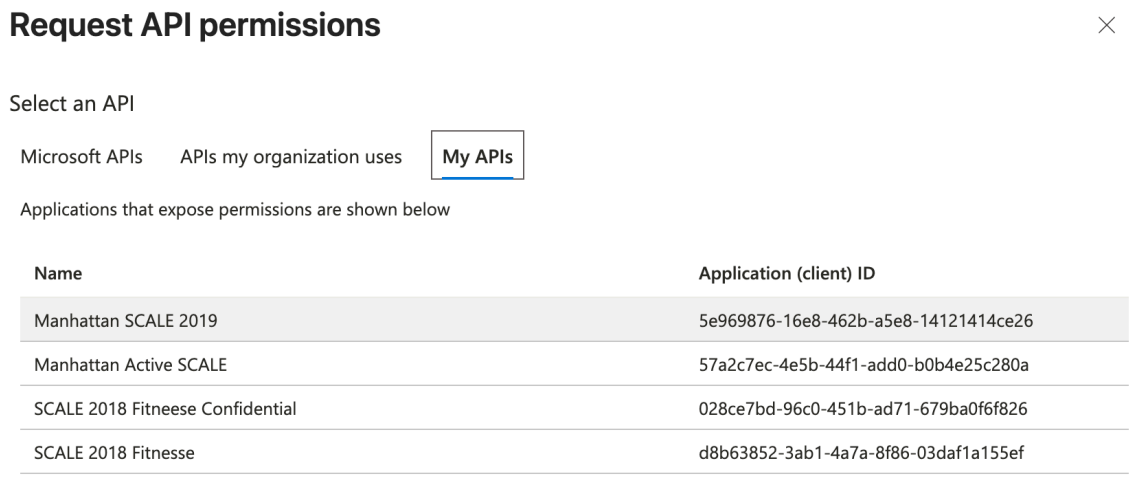
Figure 38

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



29. Select SCALE application created in the [Register SCALE as Application](#) section and click the application..

Figure 39



30. Select the **SCALEUSER** checkbox and click **Add permissions**.

Figure 40

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Request API permissions

< All APIs

MS

Manhattan SCALE 2019

What type of permissions does your application require?

Delegated permissions

Your application needs to access the API as the signed-in user.

Application permissions

Your application runs as a background service or daemon without a signed-in user.

Select permissions

expand all

Start typing a permission to filter these results

Permission	Admin consent required
Permissions (1)	
☐ Confidential ⓘ Confidential	Yes
☒ SCALEUSER ⓘ SCALEUSER	Yes

Add permissions

Discard

31. Click **Grant admin consent for directory name** and click **Yes**.

Figure 42

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

registrations > SCALE API Confidential - API permissions

permissions

Permissions have changed. Users and/or admins will have to consent even if they have already done so previously.

Applications are authorized to use APIs by requesting permissions. These permissions show up during the consent process where users are given the opportunity to grant/deny access.

[+ Add a permission](#)

API / PERMISSIONS NAME	TYPE	DESCRIPTION	ADMIN CONSENT REQUIRED
▼ Manhattan SCALE 2019 (1)			
SCALEUSER	Application	SCALEUSER	Yes ⚠ Not granted for R&D ...
▼ Microsoft Graph (1)			
User.Read	Delegated	Sign in and read user profile	-

These are the permissions that this application requests statically. You may also request user consent-able permissions dynamically through code. [See best practices for requesting permissions](#)

Grant consent

As an administrator, you can grant consent on behalf of all users in this directory. Granting admin consent for all users means that end users will not be shown a consent screen when using the application.

[Grant admin consent for R&D Scale Development](#)

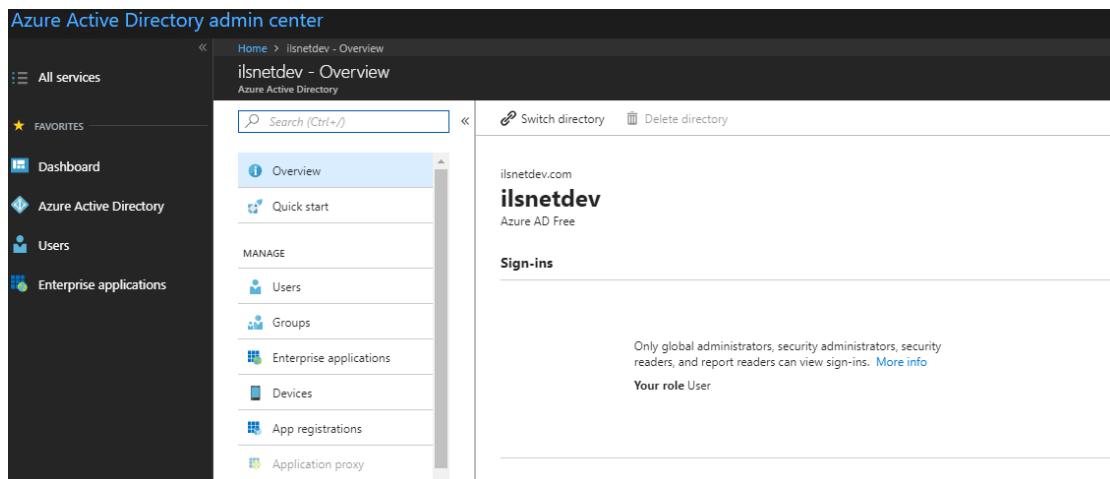
Do you want to grant consent for the requested permissions for all accounts in R&D Scale Development? This will update any existing admin consent records this application already has to match what is listed below.

[Yes](#) [No](#)

Note: Only a global administrator can grant this permission.

32. Go to the home page by changing the browser URL to <http://aad.portal.azure.com> and clicking **Enterprise applications**.

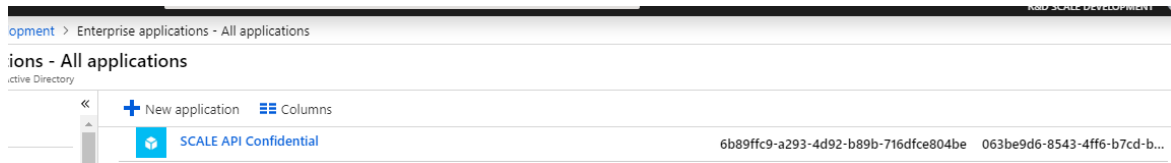
Figure 43



33. Select the application created above.

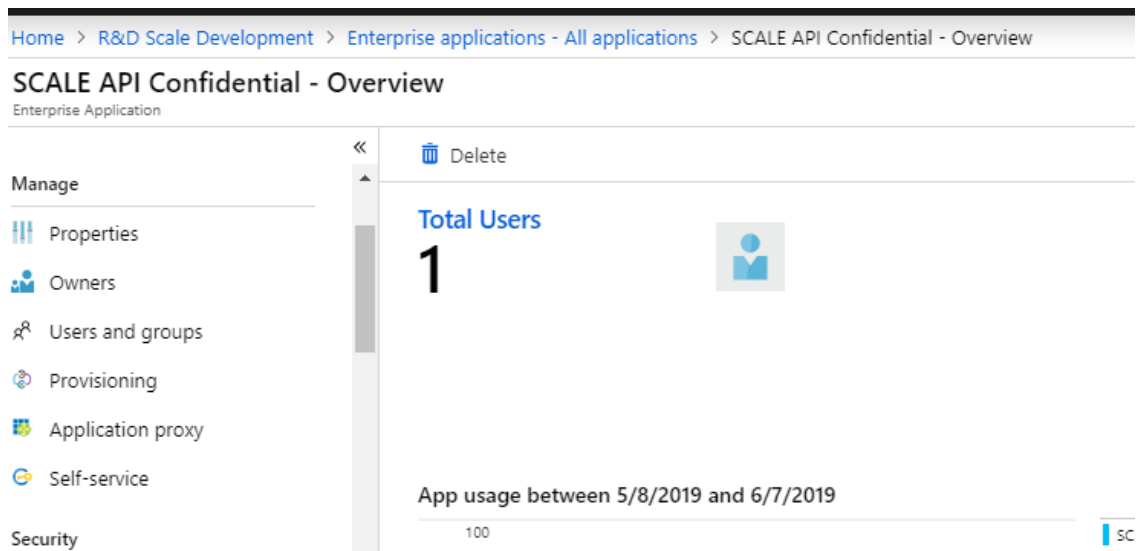
Figure 44

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



34. Click **Properties**.

Figure 45



35. Switch the **User assignment required?** option to Yes and click **Save**.

Figure 46

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

The screenshot shows the 'Properties' page for the enterprise application 'SCALEConfidentialAPP' in the Microsoft Entra ID portal. The left sidebar contains navigation links for Overview, Getting started, MANAGE (Properties, Owners, Users and groups, Provisioning, Application proxy, Self-service), SECURITY (Conditional access, Permissions), ACTIVITY (Sign-ins, Audit logs), and TROUBLESHOOTING + SUPPORT. The main content area has 'Save' and 'Discard' buttons at the top. Below them are several settings: 'Enabled for users to sign-in?' (Yes/No), 'Name' (SCALEConfidentialAPP), 'Publisher' (ilsnetdev), 'Homepage URL' (https://h-dalal-16.ilsnetdev.pri/scale), 'Logo' (SC), 'Application ID' (22f968e4-7d20-4d55-a1d6-7b39226d549f), 'Object ID' (1b996e91-b0c6-423e-a4e3-bcf0b0c7052d), 'User assignment required?' (Yes/No, highlighted with a red box), and 'Visible to users?' (Yes/No).

SCALE Application Permission

For the SCALE web application to work with Microsoft Entra ID, the global administrator of Microsoft Entra ID needs to grant permission to use this as an application.

1. Log on to the Microsoft Entra ID portal (<http://aad.portal.azure.com>)

Figure 47

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

The screenshot shows the Microsoft Entra admin center interface. The left sidebar contains navigation options: Home, Agents, Favorites, Entra ID (expanded), Overview, Users, Groups, Devices, Enterprise apps, App registrations (highlighted), Roles & admins, Delegated admin partners, Domain services, and Identity Secure Score. The main content area displays the 'R&D Scale Development' application page. It includes a search bar, tabs for Overview, Monitoring, Properties, Recommendations, and Setup guides, and a search bar for tenants. The 'Basic information' section lists details: Name (R&D Scale Development), Tenant ID (68e9df92-3a9d-4864-9e8a-85edce6c9086), Primary domain (manhdevscale.com), and License (Microsoft Entra ID Free). On the right, statistics show 215 Users, 26 Groups, 91 Applications, and 15 Devices. Alerts at the bottom indicate 9 Global Administrators and a migration to the converged Authentication methods policy.

Basic information	Value	Count
Name	R&D Scale Development	
Tenant ID	68e9df92-3a9d-4864-9e8a-85edce6c9086	
Primary domain	manhdevscale.com	
License	Microsoft Entra ID Free	
Users		215
Groups		26
Applications		91
Devices		15

- Click **App registrations** in the menu. Click **application** created [above](#)

Figure 49

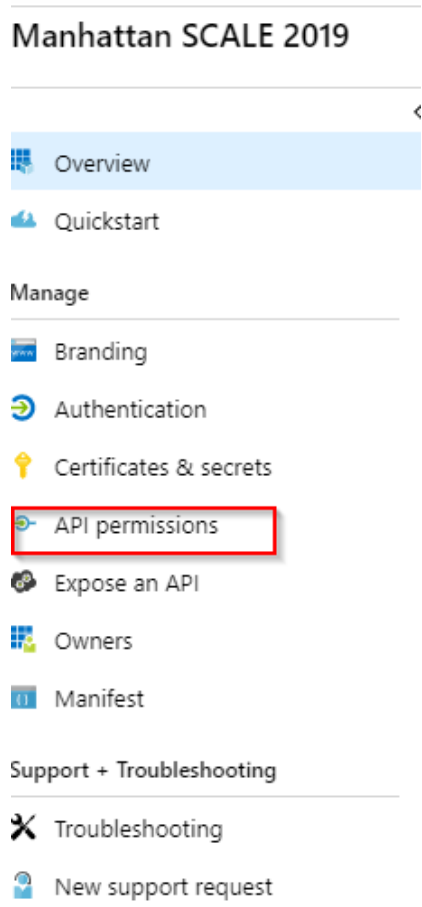
The screenshot shows the 'App registrations' page in the Microsoft Entra admin center. It includes a navigation bar with 'New registration', 'Endpoints', 'Troubleshooting', and 'Got feedback?'. A welcome message states 'Welcome to the new and improved App registrations (now Generally Available). See what's new'. A warning message indicates a change from App registrations (Legacy) and provides links to learn more or go back. The 'All applications' tab is selected, and a search bar is present. A table lists applications with columns: DISPLAY NAME, APPLICATION (CLIENT) ID, CREATED ON, and CERTIFICATES & SECRETS. The first application, 'Manhattan SCALE 2019', is highlighted with a red box.

DISPLAY NAME	APPLICATION (CLIENT) ID	CREATED ON	CERTIFICATES & SECRETS
MS Manhattan SCALE 2019	5e969876-16e8-462b-a5e8-14121414ce26	5/30/2019	-
MS Manhattan SCALE 2019 Confidential	a19f2bdd-e81f-49f4-9d24-2dcb37c53313	6/7/2019	Current
SA SCALE API Confidential	063be9d6-8543-4ff6-b7cd-b92bb3890d53	6/7/2019	-

- Select **API Permission**.

Figure 50

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



4. Click **Grant admin consent for directory name**.

Figure 51

API permissions

Applications are authorized to use APIs by requesting permissions. These permissions show up during the consent process where users are given the grant/deny access.

[+ Add a permission](#)

API / PERMISSIONS NAME	TYPE	DESCRIPTION	ADMIN
▼ Microsoft Graph (1)			
User.Read	Delegated	Sign in and read user profile	-

These are the permissions that this application requests statically. You may also request user consent-able permissions dynamically through code. [See best practices for requesting permissions](#)

Grant consent

As an administrator, you can grant consent on behalf of all users in this directory. Granting admin consent for all users means that end users will not be shown a consent screen when using the application.

[Grant admin consent for R&D Scale Development](#)

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

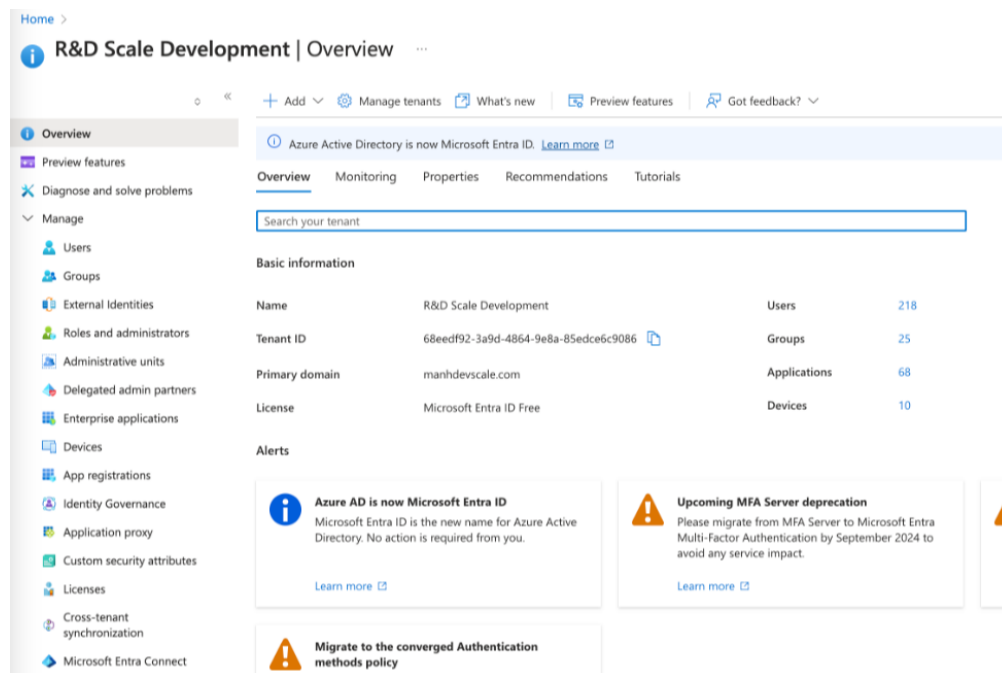
Note: Only a global administrator can grant this permission.

TPM Guest Account

External users need to be invited if they want to access TPM. Follow these steps to invite external users

1. Log on to the Azure AAD portal (<http://aad.portal.azure.com>) directly or log on to the Azure portal (<https://portal.azure.com>) and navigate to **Microsoft Entra ID**.

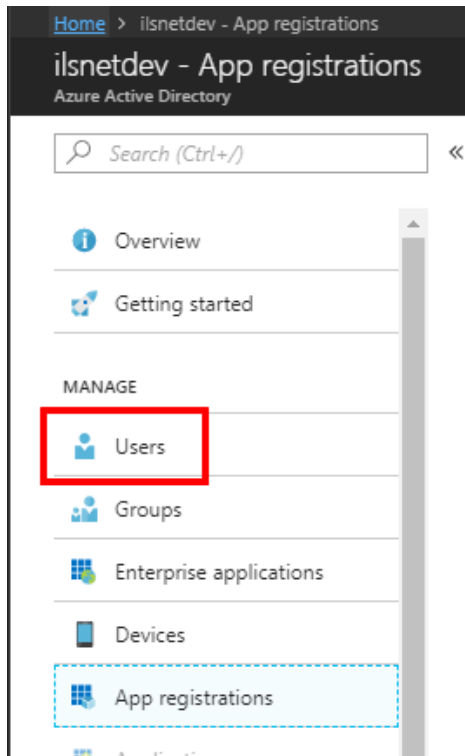
Figure 63



2. Select your directory and click **Users** in the menu.

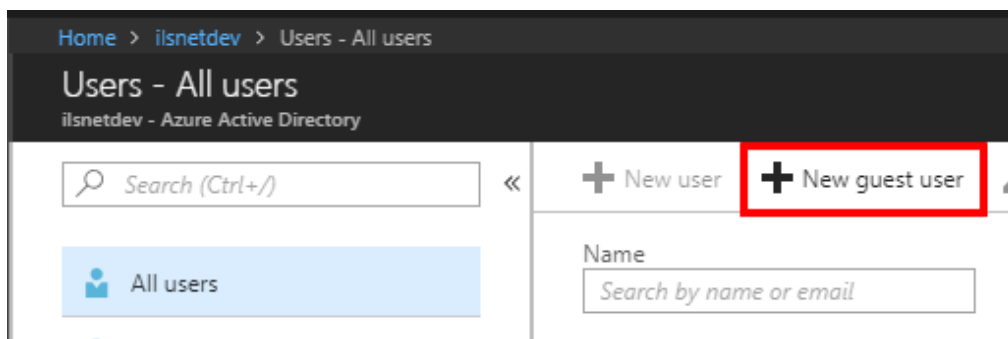
Figure 64

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



3. Click **New guest user**.

Figure 65



4. Enter the external user email ID and type a message. Click **Invite**.

Figure 67

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Home > ilsnetdev > Users - All users > New Guest User

New Guest User

ilsnetdev

i This user will be added as a Guest. Click here to learn more.

* User name ⓘ
manish.dalal2@gmail.com ✓

Include a personal message with the invitation

You are invited to access to access TPM. Please create your account and access TPM using https://<ServerName>/TPM

-Admin
Organization XYZ

Invite

5. Navigate back to **Microsoft Entra ID**. Click **Enterprise applications** and select the application created in the [Register SCALE as Application](#) section.

Figure 68

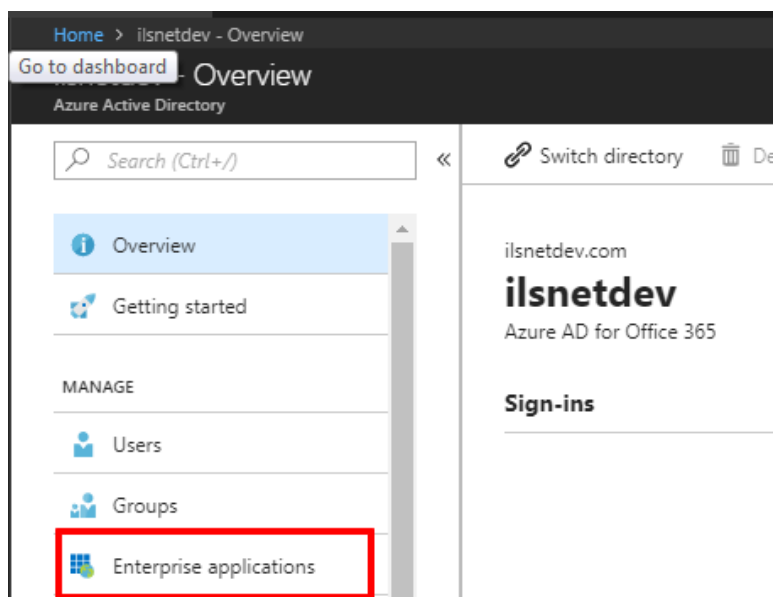


Figure 69

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

[+ New application](#) [Columns](#)

Application Type: Enterprise Applications
Applications status: Any
Application visibility: Any

First 20 shown, to search all of your applications, enter a display name or the application ID.

NAME	HOMEPAGE URL
 Manhattan SCALE	https://h-dalal-16.ilsnetdev.pri/scale

- Click **Users and groups** in the menu and then click **Add user**.

Figure 70

Home > ilsnetdev > Enterprise applications - All applications > Manhattan SCALE - Users and groups

Manhattan SCALE - Users and groups

Enterprise Application

Overview | Getting started | MANAGE | Properties | Owners | **Users and groups**

+ Add user | Edit | Remove | Update Credentials

The application will appear on the access panel for assigned users:

First 100 shown, to search all users & groups, enter a display name.

DISPLAY NAME	OBJECT TYPE
Abhilash H. Mallesha	User
Abhishek Ranjan	User
Abhishek Sengupta	User

Figure 71

Home > ilsnetdev > Enterprise applications - All applications > Manhattan SCALE - Users and groups

Manhattan SCALE - Users and groups

Enterprise Application

Overview | Getting started | MANAGE | Properties | Owners | **Users and groups**

+ Add user | Edit | Remove | Update Credentials

The application will appear on the access panel for assigned users:

First 100 shown, to search all users & groups, enter a display name.

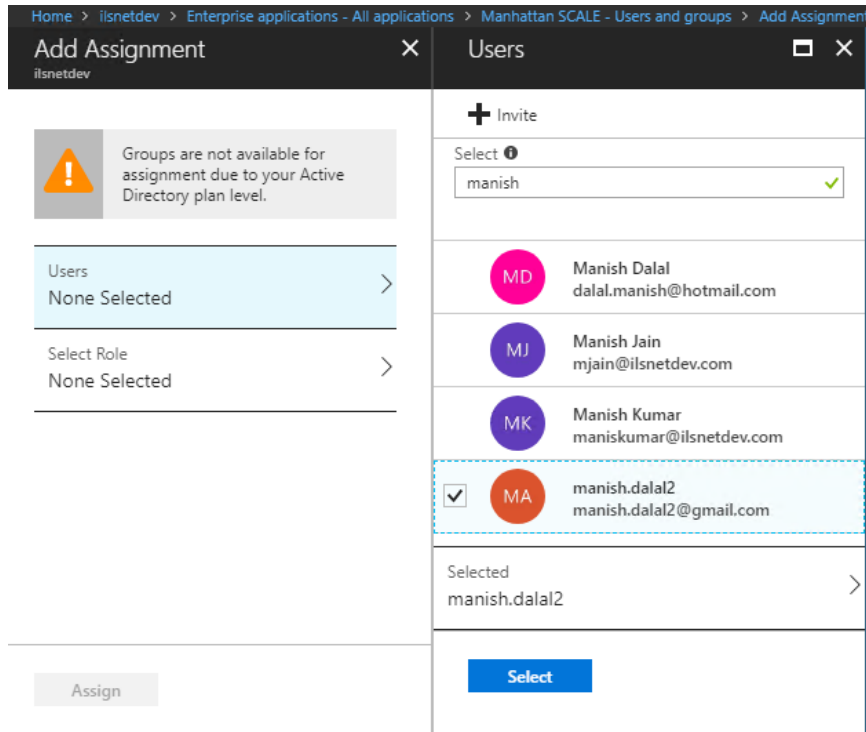
DISPLAY NAME	OBJECT TYPE
Abhilash H. Mallesha	User
Abhishek Ranjan	User
Abhishek Sengupta	User

- Click **Users** and select the users that need access to SCALE TPM application.

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

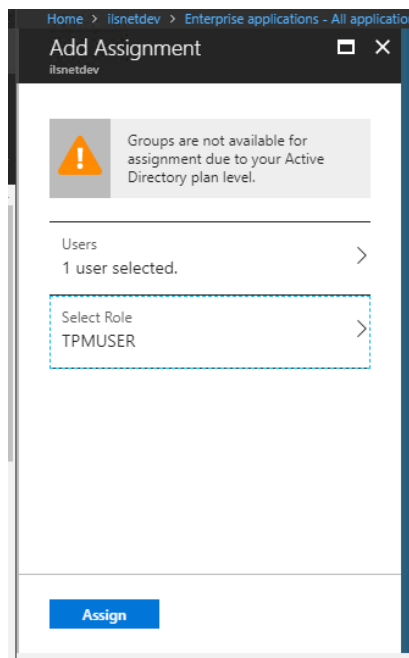
Note: If you have the basic or a higher plan of Microsoft Entra ID, you can add a group and then assign roles to it.

Figure 72



8. Click **Select Role** choose the TPMUSER role for the users and click **Assign**.

Figure 73



SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

9. The selected users will be displayed with their assigned roles in the next screen.

Figure 74

+ Add user ✎ Edit ✕ Remove 🔑 Update Credentials		
i The application will appear on the access panel for assigned users. Set 'visible to users?' to no in properties to prevent this. →		
Manish Dalal	User	Default Access
Manish Dalal	User	TPMUSER
manish.dalal2	User	TPMUSER
SCALEConfidentialAPP	Unknown	
Swale	User	TPMUSER
Swale	User	SCALEUSER
Yan Fu	User	TPMUSER
Yan Fu	User	Default Access

10. The external user will get email as below to create an account.

Figure 75

You're invited to the ilsnetdev organization

Inbox x

Microsoft Invitations <invites@microsoft.com> [Unsubscribe](#)
to me



Azure Active Directory

You've been invited to access applications in the
ilsnetdev organization
by

Manish Dalal

You are invited to access to access TPM. Please create your account and access TPM using
<https://<ServerName>/TPM>

-Admin
Organization XYZ

Get Started

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID

Configure SCALE to use Microsoft Entra ID

For SCALE to use Microsoft Entra ID, we can configure SSO using the SCALE PowerShell module.

1. Open PowerShell on SCALE application server as administrator.
2. Run command `Set-SingleSignOn`
3. Provide the values as follows:
 - **AuthenticationInstance:** <https://login.microsoftonline.com/>
 - **ClientId:** [Application Id copied here](#)
 - **Domain:** Domain of Microsoft Entra ID
 - **TenantId:** [Directory Id copied here](#)
 - **NativeClientId:** [Application Id copied here](#)
 - **ConfidentialClientId:** [Application Id copied here](#)
 - **Key:** [Secret copied here](#)
 - **APIConfidentialClientId:** [Application Id copied here](#)
 - **EnvironmentName:** This is the name of environment you want to update this SSO value of. Empty string means Default Database.

Note: Application Instance for China should be **sts.xyz.com** instead of **login.xyz.com**.

Figure 76

```
PS C:\Windows\system32> Import-Module "C:\Program Files\Manhattan Associates\ILS\2018\Toolbox\Powershell\SCALEPowerShell.dll"
PS C:\Windows\system32> Set-SingleSignOn

cmdlet Set-SingleSignOn at command pipeline position 1
Supply values for the following parameters:
AuthenticationInstance: https://login.microsoftonline.com/
ClientId: 725e5293-a539-4194-a32a-fcb387
Domain: ilsnetdev.com
TenantId: e425df28-d707-429a-ab3a-c820d52e3ad8
NativeClientId: 725e5293-a539-4194-a32a-fcb387
ConfidentialClientId: 22f968e4-7d20-4d55-a1d6-7b39226d549f
Key: g2xRZhUQgr3argIEaI+vVS4TF25958tPY3AC1qWnhns=
EnvironmentName:
PS C:\Windows\system32>
```

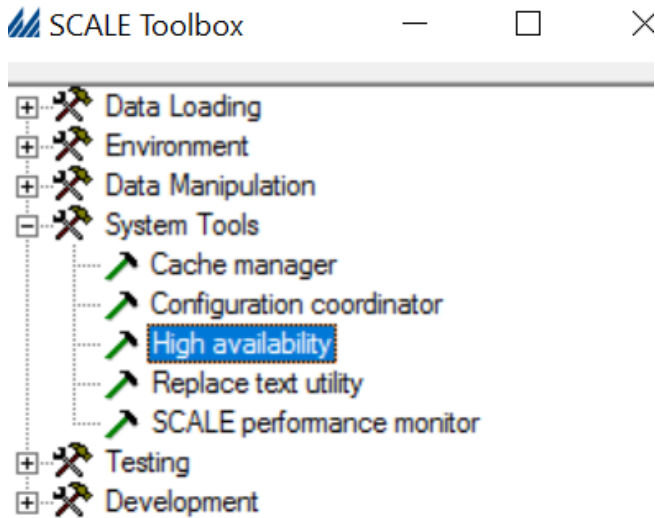
Note: Once we update the SSO values it is mandatory to do ILS Reset (proceed to C:\Program Files\Manhattan Associates\ILS\<version>\Utilities and then click on ILS Reset Services) for the values to be reflected.

Configure load balancer

1. Open the SCALE toolbox as admin.
2. Navigate to System Tools and double-click click High Availability option

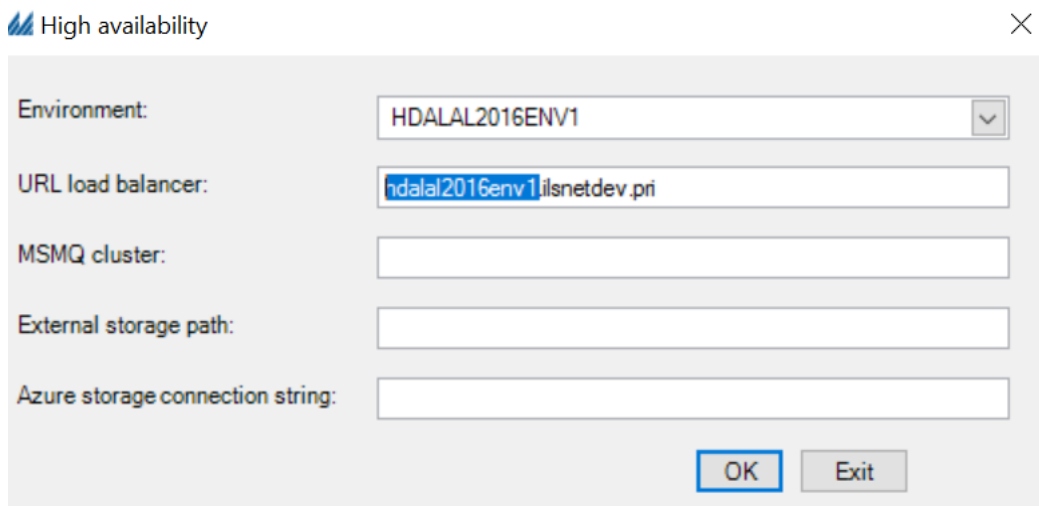
Figure 77

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



3. Update the URL load balancer as discussed above ([Without SCALE environment/](#) [With SCALE environment](#))

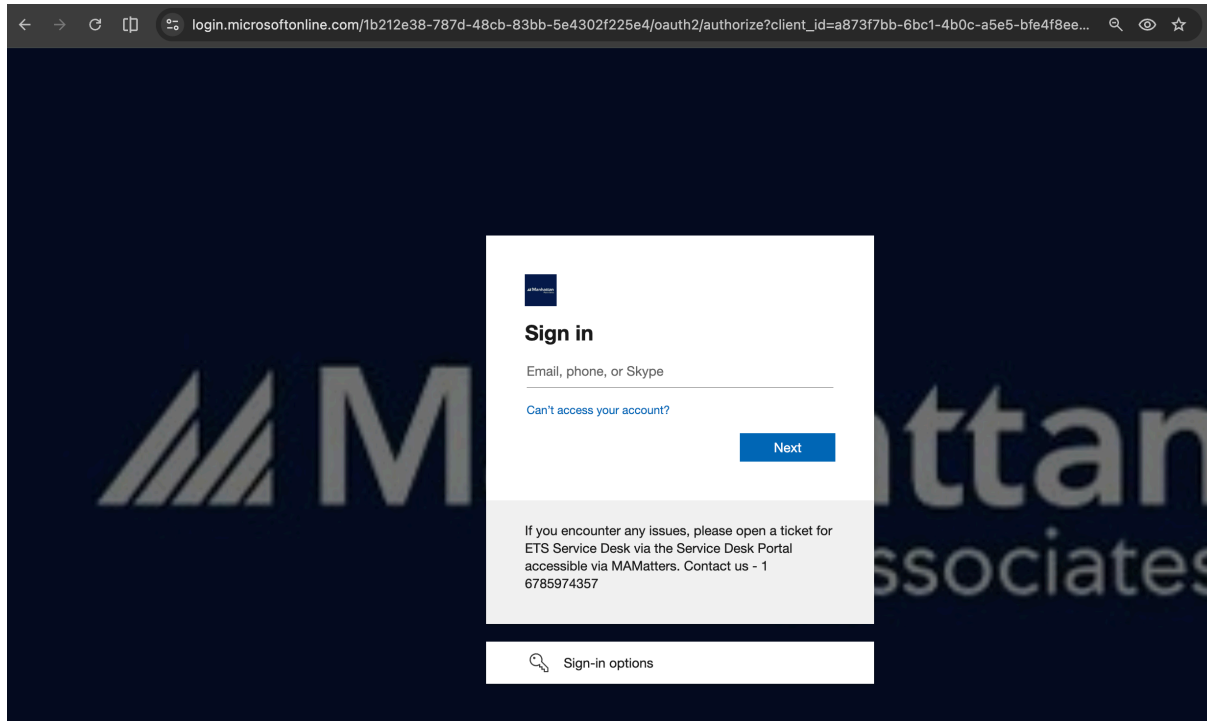
Figure 78



Now open a browser and navigate to the SCALE application using the full domain name (i.e. <https://hostname/SCALE/security/login>. You will be redirected to <https://login.microsoftonline.com> and asked for your domain login id and password if not logged in with your domain account already.

Figure 79

SINGLE SIGN-ON GUIDE: MICROSOFT ENTRA ID



Upgrading customer

Customer Upgrading from 2021/2022 to 2026

If you are upgrading from SCALE 2021/2022 to 2026, please follow step 12 for CONFIGUSER in section [Register SCALE as Application](#) section.